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MATHEMATICS II
TEACHING MODULE
FORM 2
TERM 2



Developed by the Curriculum Development Centre
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MATHEMATICS II
TEACHING MODULE
FORM 2
TERM 2

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Vision

Quality, life- long education for all which is accessible, inclusive and relevant to individual, national and global needs and value system.

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Preface

The Mathematics II Teaching Module for Form II is designed to help teachers prepare and deliver Competence-Based lessons with a comprehensive understanding of Mathematics concepts, fostering a deep appreciation for the role of Mathematics in everyday life and its applications in various fields. This module aims to assist teachers to help learners develop analytical thinking, problem-solving skills, and project based skills through a structured and progressive learning approach. This module prescribes on hands-on activities, inquiry-based learning experience, encouraging learners to explore, experiment, and engage in problem solving reasoning.

This Mathematics II Teaching Module for Form 2 aims to help teachers create a stimulating and supportive learning environment where learners can develop a profound understanding of Mathematics. This module will help teachers to guide learners to grow intellectually and personally by preparing them for professions in Science, Technology, Engineering and Mathematics (STEM) as well as for higher education by encouraging curiosity, critical thinking, and practical skills.

It is hoped that this module will inspire learners to explore the fascinating world of Mathematics and appreciate its significance in shaping the future.

Kelvin Mambwe, (PhD.)
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Last but not the least, the commitment and hard work of all the staff at the Curriculum Development Centre in ensuring that this module comes to reality is recognised.

Charles Ndakala, (Dr.)
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HOW TO USE THIS MODULE

To effectively use this Mathematics II Teaching Module for Form 2:

- Read and familiarise yourself with the module's content, learning activities, and assessment guides.
- Plan your lessons in advance, using the module's suggested teaching and learning activities.
- Use a variety of teaching methods, including demonstrations, discussions, group work, and hands-on activities.
- Encourage active learning by asking open-ended questions, promoting critical thinking, and fostering problem-solving skills.
- Assess learners learning regularly, using the module's suggested assessment strategies and tools.
- Provide feedback and support to learners, helping them to identify areas for improvement and develop their skills.
- Integrate technology into your teaching, using multimedia resources and interactive simulations to enhance learner engagement and understanding.
- Monitor learners progress and adjust your teaching strategies as needed, to ensure that all learners meet the learning objectives

By following these steps, you can effectively use this Mathematics II Teaching Module for Form 2 to support your teaching and promote learning.

INTRODUCTION

Mathematics, as a fundamental STEM subject, which plays an important role in shaping learners' understanding of the natural world. As learners transition from upper primary to secondary education, it is necessary to provide them with an all-inclusive and engaging introduction to Mathematics. Therefore, the Mathematics II Teaching Module for Form 2 is a comprehensive educational resource designed to support the teaching and learning of Mathematics for learners in their first year of secondary education. It covers fundamental concepts and principles of Mathematics, aligning with the 2024 Mathematics syllabus. The module includes topic and subtopic overviews, suggested teaching and learning materials, environment set-up, learning activities, and assessment guides. Additionally, it incorporates teaching methods and strategies like inquiry-based learning, problem-solving activities, group discussions, hands-on activities, multimedia resources, and interactive simulations. The aim is to provide a solid foundation in Mathematics, foster critical thinking and problem-solving skills, and inspire further studies in Mathematics.

SUGGESTED TEACHING AND LEARNING MATERIALS

The study of Mathematics content requires hands-on experiences, visual aids, and interactive resources to strengthen deep understanding and appreciation. To support the teaching and learning of Mathematics II in Form 2, the suggested teaching and learning materials are meant to:

- Enhance learner engagement and motivation in Mathematics
- Develop practical skills and inquiry based thinking
- Promote critical thinking, problem-solving, and analytical skills
- Support differentiated instruction and inclusive learning

The suggested teaching and learning materials are either artificial or natural. By utilising these suggested teaching and learning materials, teachers can create a dynamic and supportive learning environment that promotes academic excellence, creativity, and scientific literacy in Mathematics II for Form 2 learners.

LEARNING ENVIRONMENT SET UP

To create an effective learning environment for teaching and learning Mathematics is important in deepening understanding of the concepts and application in real life context. The learning environment set-up aims to create a safe, inclusive, and engaging space.

- **Natural Environment:** A natural learning environment is a setting where learners explore and learn naturally, often without explicit instruction or formal teaching, such as in school surroundings
- **Artificial Environment (Man-Made):** Artificial learning environments are intentionally designed safe spaces, such as classrooms, laboratories, and libraries, designed for formal instruction, hands-on activities, games, and songs

- **Technological Learning Environment:** Access educational apps, games, and software for learning, including game-based platforms, virtual platforms, and simulations, to engage learners and promote learning

SAFETY IN THE LEARNING ENVIRONMENT

Safety in the learning environment is a requirement in learning Mathematics, as learners are exposed to potential hazards during hands-on activities. Collaboration between teachers, and learners, is important to create a responsible learning environment that promotes scientific inquiry. Guidelines for maintaining a safe learning environment include, risk identification, personal protective equipment, emergency response plans, storage, disposal, and learner responsibilities. Prioritising safety minimises risks, prevents accidents, and ensures a positive learning experience.

TEACHING METHODOLOGY

The effective teaching methodologies in Mathematics II include:

- **Conducting projects:** Demonstrate key principles and encourage curiosity among learners.
- **Collaborative learning:** Pair learners to work together, promoting peer-to-peer teaching, discussion, and problem-solving.
- **Conceptual learning:** Connect mathematical concepts to everyday life, industry, or current events, making learning relevant and meaningful.
- **Differentiated instructions:** Tailor teaching to meet diverse learning styles, abilities, and interests of different learners.
- **Feedback and reflection:** Encourage learners to reflect on their learning, providing constructive feedback to guide improvement.
- **Inquiry-based learning:** Encourage learners to explore, investigate, and discover Mathematical concepts through hands-on activities.
- **Integration of technology:** Use digital tools, simulations, and visualisations to enhance engagement, understanding, and analysis.
- **Problem-based learning:** Present real-world problems or case studies or scenarios, requiring learners to apply Mathematical knowledge to develop solutions.
- **Project -based learning:** Assign open-ended projects, allowing learners to design, conduct, and present research or applications of Mathematical concepts.

By implementing these methodologies, a teacher can create an engaging, inclusive, and effective Mathematics learning environment.

LEARNING ACTIVITIES

Learning activities are intentional educational experiences aimed at promoting learning, engagement, and achievement among learners. Facilitated by teachers, they help acquire new knowledge, skills, attitudes, and behaviour change. To create an inclusive environment, teachers should use a "hook" or problem posing or scenario or key question or case studies to introduce new learning activities in an interactive and interesting way.

ICONS USED IN THIS MODULE

This module utilises icons as visual symbols or graphics to represent instructions, enhancing the learning experience for learners. Icons categorise and organise instructions, making navigation easier for teachers.



Key Terms



Assessment



Problem Posing



Activity



Tips



Study skills



Summary

TIME ALLOCATION

The standard minimum learner-teacher contact time for Mathematics II at secondary school level is 4 hours per week, translating to Six (6) periods with at least two double periods per week. The duration for a single period is 40 minutes. The contact time at Secondary school level is planned in such a way as to give ample time for practical activities.

ASSESSMENT

Formative Assessments: To monitor learner progress, identify areas of improvement, and adjust instruction to meet learner needs.

Quizzes: Regular quizzes to assess learners' understanding of concepts.

Tests: Periodic tests to evaluate learners' knowledge and application of Mathematics concepts.

Class Discussions: Observing learners' participation and engagement in class discussions.

Group Work: Assessing learners' ability to work collaboratively and contribute to group tasks.

Projects: Evaluating learners' ability to design, conduct, and present a Mathematics project.

Presentations: Assessing learners' ability to communicate Mathematics concepts and ideas effectively

Summative Assessments: To evaluate learners' learning at the end of a lesson, sub-topic, topic or term, and to provide a comprehensive picture of learner achievement.

Unit Tests: Comprehensive tests to evaluate learners' understanding of Mathematics concepts at the end of each unit.

Mock Exams: Comprehensive exams to evaluate learners' overall understanding of Mathematics concepts at the end of the Term.

Practical Tasks: Assessing learners' problem solving skills and techniques through practical exams.

Project-Based Assessments: Evaluating learners' ability to design, conduct, and present a Mathematics project.

Key Competences

COMPETENCE	DESCRIPTORS
Analytical Thinking	• Identify patterns • Compile data, create mental images and address issues • Evaluate solutions
Collaboration	• Solving puzzle in groups • Play with peers to build relationships • Participate in and express themselves through play activities
Communication	• Use mathematical/scientific language in different situations • Express oneself using different media and symbols • Ask for feedback
Creativity and Innovation	• Apply knowledge and skills in new and creative ways • Express ones' ideas in logical manner • Use different strategies and approaches in response to a problem or question • Explain complex ideas
Critical Thinking	• Ask and answer simple questions • Classify objects according to their attributes • Manipulate different objects • Solve simple problems in life • Match different things according attributes • Arrange objects according to attributes • Compare similarities or differences between objects • Explore the environment • Differentiate good from bad • Recognise and name items in the environment
Digital Literacy	• Perform successful online research • Manage digital projects by effective planning and executing • Work with data through data analysis, visualisation, and data base management • Produce graphs for data analysis
Problem Solving	• Make connections/link with the inner world or social environment • Use numeracy patterns and relations to solve problems • Manipulate numbers, shapes and symbols to complete a task

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TOPIC 4: ALGEBRA

Introduction

Overview: Algebra is a branch of Mathematics that uses letters and symbols to represent numbers and relationships between numbers. It helps us to represent unknown values using letters, write mathematical statements called expressions and equations, find unknown values by solving equations, and describe patterns and real-life situations mathematically. At Form 2 level, algebra builds on the foundational ideas introduced in Form 1 and focuses on helping learners understand and use Algebra as a tool for representing and solving real-life problems. Through learner-centred activities, discussions, and practical tasks, learners develop confidence in manipulating algebraic expressions and applying algebraic methods accurately.

This topic has two sub-topics namely basic processes, and remainder and factor theorems.

General Competences

- ☐ **Analytical Thinking:** Break down word problems into algebraic statements, interpret algebraic expressions, identify relationship between variables, and compare different solution methods and check accuracy of results.
- ☐ **Collaboration:** Work in groups to solve problems on algebra and compare solutions.
- ☐ **Communication:** Explain algebraic processes clearly using Mathematical language.
- ☐ **Creativity and Innovation:** Use Algebra to model real-life situations in different ways, create patterns and general rules using algebraic symbols, and apply algebra in unfamiliar or practical contexts such as finance, measurements, and patterns.
- ☐ **Critical Thinking:** Justify answers logically and mathematically, detect errors in worked examples, correct them, and make informed decisions based on algebraic reasoning.
- ☐ **Digital Literacy:** Use calculators or digital tools to check algebraic solutions.
- ☐ **Problem solving:** Solve problems involving basic processes, remainder and factor theorems.



Key Terms/Vocabulary:

- ☐ **Algebra:** A branch of Mathematics that uses letters and symbols to represent numbers
- ☐ **Dividend:** The polynomial being divided.

- **Divisor:** The expression by which a polynomial is divided
- **Factor:** An expression that divides the polynomial exactly (remainder = 0)
- **Linear factor:** A factor of degree 1, usually written as: $(x - a)$
- **Polynomial:** An algebraic expression made up of numbers and variables raised to whole number powers and connected by addition or subtraction.
- **Quadratic expression:** This is a polynomial of degree 2.
- **Quotient:** The result obtained after division (excluding the remainder)
- **Remainder:** The value left after dividing one polynomial by another
- **Zero (Root):** A value of x that makes $P(x) = 0$

Sub-Topic 1: Basic Processes

Introduction

In Form 2, learners are expected to deal with basic processes of Algebra, such as factorising by grouping terms, quadratic expression, and difference of two squares. They are also expected to add, subtract, multiply and divide algebraic fractions, and simplify algebraic fractions using the Lowest Common Multiples (LCM) and factorisation. These basic processes help learners think logically, solve problems, and apply Mathematics to real-life situations. Algebra also provides a foundation for more advanced topics in Mathematics.

Specific Competence: Apply basic processes of Algebra in different contexts



Learning Activity 1: Exploring patterns and relationships in relation to Algebra

Introduction

Algebra is the language we use to describe patterns and relationships in everyday life. From the number of tiles in a growing design, the cost of buying several items, or the distance travelled over time, patterns help us to predict what comes next. In this unit, it is important to let learners explore number and visual patterns, identify relationships between quantities, and express these relationships using algebraic expressions and simple formulae. By recognising how quantities change together, learners develop the foundation for understanding variables, equations, and functions. Through hands-on investigations, discussions, and real-life problem-solving, learners

will move from observing patterns to representing them symbolically using letters and algebraic expressions.



Problem Posing: A school is building a square courtyard using tiles. Each stage adds one new square boarder of tiles around the previous one. The number of tiles added at each stage follows this pattern:

Stage (n)	1	2	3	4
Tiles Added	5	9	13	17

- Find the number of tiles added in stage 5 and stage 10,
- Let $T(n)$ represent the number of tiles added in stage n . Find an algebraic rule for $T(n)$.
- Let $S(n)$ represent the total number of tiles used after n stages. Find a formula for $S(n)$.
- If the school has 500 tiles, determine the largest stage that can be completed.

Activity 1.1: Research on patterns and relationships in relation to Algebra

- ☐ **Artificial environment:** Books with content on patterns and relationships in relation to Algebra or any other research materials.
- ☐ **Technological environment:** Computers/phones/tablets with internet connectivity, and a projector for presentations.

What to do: Put learners in small manageable groups of mixed abilities. Provide them with computers/phones/tablets or books with content on patterns and relationships in relation to Algebra. Ensure each learner participates in the activity.

Task: Ask learners in groups to explore the patterns and relationships in relation to Algebra. Let them present their findings to the class.



Skills to be developed: Collaboration, Communication, Critical thinking, Presentation, ICT, and Research skills.

Activity 1.2: Matchstick pattern investigation

Suggested teaching and learning materials to set up the learning environment.

- ☐ **Natural environment:** Sticks
- ☐ **Artificial environment:** Matchsticks or toothpicks

What to do: Start this activity with a problem posed. Divide learners into small manageable groups of mixed abilities. Provide sticks or matchsticks or toothpicks to each group. Ensure that every learner participates in group work activity, and provide support to learners where necessary.

Task: Ask learners in each group to build growing shapes, such as:

1. Build one (1) square using matchsticks
2. Build two (2) connected squares
3. Build three (3) connected squares

Let learners count the number of matchsticks each time and record in a table like the one below.

Number of Squares (n)	Number of Matchsticks /Toothpicks/Sticks (m)
1	4
2	7
3	10
4	?

Guide them to identify the pattern (+3 each time), and develop a formula: $m = 3n + 1$



Skills to be developed: Critical thinking, Collaboration, Communication, Problem-solving.

Conceptualisation: Make sure that the concepts of patterns and relationships in relation to Algebra come out clearly. Let them know that relationships in Algebra refer to the connections between variables and quantities. These relationships can be expressed using tables, graphs, words or symbolic expressions.

Synthesis: Connect exploring of patterns and relationships in relation to Algebra to real-life situations.

Evaluation: Check if learners can explore patterns and relationships in relation to Algebra in real life. Use a checklist to check the understanding of learners.



Assessment: Give learners a project-based task to confirm that they can explore patterns and relationships in relation to Algebra and apply them in real-life situations.



Learning Activity 2: Factorising by grouping terms, quadratic expressions, and difference of two squares

Introduction

Factorising by grouping terms, quadratic expressions, and the difference of two squares are important skills in Algebra because they help simplify expressions, solve equations, and develop deeper mathematical understanding. Mastering these techniques strengthens learners' ability to recognise patterns and relationships, which are central to success in Algebra and higher-level mathematics.



Problem Posing: A digital safe will only open if you completely factor the expression shown on its screen. One day it showed the expression $x^3 + 3x^2 - 4x - 12$ on the screen. To open this safe the following procedure should be followed:

1. Use factorisation by grouping terms,
2. Obtain a quadratic expression factor,
3. Then factor it further using the difference of two squares.

Mr. Phiri has a challenge opening the digital safe. As learners of Mathematics II, help Mr. Phiri to open the digital safe.

Activity 2.1: Factorising by grouping terms

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Worksheets containing questions to factorise by grouping terms
- ☐ **Technological environment:** Computers/phones/tablets with a video on factorisation by grouping terms

What to do: Start with the problem posed. Put learners in small manageable groups of mixed abilities. Provide them with computers/phones/tablets with downloaded video on factorisation by grouping terms. Give them instructions on how to find and play the video on the gadgets provided. Thereafter, provide them with worksheets containing questions to be factorised by grouping terms. Ensure that every learner participates in group work activity.

Task: Ask learners in groups to factorise the questions by grouping terms, and present their findings to the class.



Skills to be developed: Collaboration, Communication, Critical thinking, Digital literacy, and Presentation

Activity 2.2: Factor pair investigation (Factorising quadratic expressions)

Suggested teaching and learning materials to set up the learning environment.

- ☐ **Artificial environment:** Worksheets containing quadratic expressions and the table as shown below.

What to do: Put learners in small manageable groups of mixed abilities. Provide them with worksheets containing quadratic expressions to be factorised (e.g., $x^2 + 7x + 12$, $2x^2 - x - 6$). Make sure that every learner participates in the activity.

Task: Let learners in groups list the factors, identify pair sums to the given sums in the expressions, and write a factorised form of the algebraic expression. Let them complete the table shown below.

Expression	Product	Sum	Factors	Factorised Form
$x^2 + 7x + 12$	12	7		
$2x^2 - x - 6$	-12			



Content Tip: You can also assist learners by having factors expressed as $(x + \dots)(\dots + 4)$ and ask the learners to fill in the missing variable or constant.



Skills to be developed: Collaboration, Communication, Critical Thinking, and Problem Solving

Activity 2.3: Difference of two squares (Paper Folding Model)

Suggested teaching and learning materials to set up the learning environment.

- **Artificial Environment:** Plain papers, scissors, ruler

What to do: Put learners in small manageable groups of mixed abilities. Give each group a plain paper and a scissors for cutting. Ensure that every learner participates in the activity, and provide support where necessary. Make sure that x units is longer than 4 units.

Task: Ask learners in their respective groups to draw a square of side x on a plain paper. Let them cut out a smaller square of side 4 units and calculate the large area $= x^2$. Let them also calculate the small area $= 16$. Ask them to express the remaining area, and rearrange the pieces to form a rectangle.



Content tip: Let learners discover $x^2 - 16 = (x - 4)(x + 4)$, and expose them to a variety of questions on factorisation using the difference of two squares. You can also expose learners to questions where they start by factorising by common factor and then difference of two squares.



Skills to be developed: Collaboration, Communication, and Critical thinking.

Conceptualisation: Make sure that the concepts of factorisation by grouping terms, quadratic expressions, and difference of two squares come out clearly, and give brief notes on them.

Synthesis: Connect factorising by grouping terms, quadratic expressions, and difference of two squares to real-life situations.

Evaluation: Check if the learners can factorise expressions by grouping terms, quadratic expressions, and difference of two squares and relate the concepts to real-life situations. Use a checklist to check the understanding of learners.



Assessment: Give learners tasks to confirm that they can factorise expressions by grouping terms, quadratic expressions, and by difference of two squares, and apply them in real-life situations



Learning Activity 3: Adding, subtracting, multiplying, and dividing algebraic fractions

Introduction

Algebraic fractions are fractions in which the numerator, the denominator, or both contain algebraic expressions instead of just numbers. Like numerical fractions, algebraic fractions can be added, subtracted, multiplied, and divided, but they require additional skills such as factorisation, identifying restrictions, and finding common denominators. Understanding operations on algebraic fractions help learners to simplify complex expressions, solve algebraic equations, model real-life situations (such as, rates, ratios, speed, density, financial formulas), and build a foundation for calculus and advanced algebra.



Problem Posing: Learners at a certain secondary school are preparing juice for a school event. The first group produces $\frac{2}{x}$ litres of juice. The second group adds $\frac{3}{x+1}$ litres to the mix. During serving, $\frac{1}{x}$ litres of juice spill. Finally, the remaining juice is shared equally among x learners. Write down an algebraic expression representing the amount of juice each learner receives and simplify it as much as possible.

Activity 3.1: Adding and subtracting algebraic fractions.

Suggested teaching and learning materials to set up the learning environment.

- ☐ **Artificial environment:** Worksheets containing algebraic fractions to be added and subtracted.

What to do: Put learners in small manageable groups of mixed abilities. Provide them with worksheets containing algebraic fractions. Assist the learners where necessary and ensure that every learner participates fully.

Task: Ask learners in their respective groups to add and subtract algebraic fractions, and present their findings to the class.



Content tip: Ensure that a variety of questions such as $\frac{5}{x-2} - \frac{3}{x+4}, \frac{1}{2x+3} + \frac{2}{x-1}$ and $\frac{3}{x^2-9} + \frac{2}{x+3}, \frac{5}{x^2-x-6} - \frac{1}{x-3}$ are covered.



Skills to be developed: Collaboration, Communication, Critical thinking, and Problem solving.

Activity 3.2: Multiplying and dividing algebraic fractions

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Worksheets with questions on multiplication and division of algebraic fractions.
- ☐ **Technological environment:** Computers/phones/tablets with a video on multiplication and division of algebraic fractions.

What to do: Put learners in small manageable groups of mixed abilities. Provide them with computers/phones/tablets with downloaded video on multiplication and division of algebraic fractions. Give them instructions on how to find and play the video on the gadgets provided. Thereafter, provide them with worksheets containing questions to be multiplied and divided. Ensure that every learner participates in group work activity.

Task: Ask learners in their respective groups to multiply and divide algebraic fractions, and present their findings to the class.



Content Tip: Ensure that a variety of questions such as $\frac{x^4y^3}{y^2} \times \frac{x^2y^2}{xy^4}, \frac{15k^2}{7cd^2} \div \frac{5k^3b^2}{21c^2d^3}, \frac{x^2y^3}{2z} \times \frac{xz^2}{2y} \div \frac{y^2z}{2x}, \frac{x^2-9}{x^2-3x} \div \frac{2x}{x-3}, \frac{x^2-9}{x^2-4} \times \frac{x-2}{x+3}$ and many others.



Skills to be developed: Collaboration, Communication, Critical thinking, Presentation, and Problem solving.

Conceptualisation: Make sure that learners can add, subtract, multiply and divide algebraic fractions correctly, and give them brief notes.

Synthesis: Connect adding, subtracting, multiplying, and dividing algebraic fractions to real-life situations.

Evaluation: Check if the learners can add, subtract, multiply, and divide algebraic fractions and connect them to real life situations. Use a checklist to check the understanding of learners.



Assessment: Give learners exercises to confirm that they can add, subtract, multiply and divide algebraic fractions and apply them in real-life situations.



Learning Activity 4: Simplifying algebraic fractions using Lowest Common Multiples (LCM) and factorisation

Introduction

Simplifying algebraic fractions using Lowest Common Multiples (LCM) and factorisation helps learners transform complex expressions into simpler, more manageable forms. By mastering factorisation and understanding how to find the LCM, learners gain strong foundational skills necessary for solving equations, working with rational expressions, and tackling advanced algebra. This learning activity builds both procedural fluency and deep conceptual understanding, as well as essential skills for success in secondary Mathematics.



Problem Posing: Learners at a secondary school are fundraising some money to construct a laboratory. Group A raises $\frac{x-2}{x^2-4}$ of the total funds. Group B raises $\frac{2x+4}{x^2+3x+2}$ of the total funds. The organisers want to combine the contributions to know the total funds raised. Simplify the expression representing the total funds raised.

Activity 4.1: Simplifying Algebraic Fractions

Suggested teaching and learning materials to set up the learning environment.

- **Artificial environment:** Worksheets containing algebraic fractions to be simplified.

What to do: Put learners in small manageable groups of mixed abilities. Start with the problem posed. Provide the groups with worksheets containing different algebraic fractions to be simplified. Ensure that all learners participate in group work activity. Provide support to learners as they work in groups.

Task: Ask each group to simplify algebraic fractions and present their findings to the class.



Content tip: Ensure that learners are exposed to a variety of questions such as

$$\frac{x-2}{x^3-4x}, \frac{x^2-16}{x+4}, \frac{p^2-q^2}{p-q}, \frac{3x^2+6x}{8x^2+16x}, \frac{2x^2-6x}{x^2-5x+6} \text{ and many others.}$$



Skills to be developed: Collaboration, Communication, Critical thinking, and Problem-solving

Conceptualisation: Make sure that learners can simplify algebraic fractions using Lowest Common Multiples (LCM) and factorisation.

Synthesis: Connect simplifying algebraic fractions using LCM and factorisation to real-life situations.

Evaluation: Check if the learners can simplify algebraic fractions using LCM and factorisation. Use a checklist to check learners' understanding.



Assessment: Give learners tasks to confirm that they can simplify algebraic fractions using LCM and factorisation, and apply them in real-life situations.

Expected Standard: Basic processes of algebra applied accurately in different contexts

Sub-Topic 2: Remainder and Factor Theorems

Introduction

The remainder and factor theorems are important concepts in Algebra because they help learners to understand the relationship between polynomial division and factorisation. These theorems provide a quicker and more meaningful way of determining the remainder when a polynomial is divided by a linear expression, and of checking whether a given expression is a factor of a polynomial. Through this topic, learners develop the ability to simplify polynomial work, factorise cubic expressions, and solve cubic as well as identical cubic equations in an efficient and logical manner.

The study of the remainder and factor theorems also strengthens learners' reasoning, analytical thinking, and problem-solving skills. It enables them to see patterns in algebraic expressions and apply algebraic ideas to both classroom and real-life situations. In this topic, learners will explore the remainder and factor theorems, use them to find quotient and remainder, factorise cubic expressions, solve cubic equations, and use ICT tools to enhance understanding.



Learning Activity 5: Exploring the Remainder and Factor Theorems

Introduction

Instead of performing lengthy polynomial division every time, mathematicians use two powerful tools, namely the Remainder Theorem and the Factor Theorem. These theorems provide efficient shortcuts that connect algebraic division with simple substitution, thereby transforming a complex procedure into a quick calculation. This section explores how and why these theorems work, helping learners move from procedural understanding to conceptual mastery.



Problem Posing: A factory machine produces packets in groups following a pattern given by the rule $P(x) = x^3 - 3x^2 + 2x + 4$. One day the manager sets the machine to $x = 2$, but instead of producing perfect packets, some items are left over. How can we predict the leftover items without actually running the machine? Is there a faster way than dividing the whole expression?

Activity 5.1: Research on the Remainder and Factor Theorems

Suggested teaching and learning materials to set up the learning environment.

- ☐ **Artificial environment:** Books or any study materials with content on the remainder and factor theorems.
- ☐ **Technological environment:** Computer/phones/tablets with internet connectivity, and a projector for presentations.

What to do: Put learners in small manageable groups of mixed abilities. Provide them with computer/phones/tablets or books/ any study materials with content on remainder and factor theorems.

Task: Ask learners to research on the remainder and factor theorems, and present their findings to the class.



Skills to be developed: Collaboration, Communication, Digital literacy, Presentation, and Research skills.

Conceptualisation: Make sure that learners can explore the remainder and factor theorems correctly, and give them brief notes.

Synthesis: Connect the remainder and factor theorems to real-life situations.

Evaluation: Check if the learners can explore the remainder and factor theorem. Use a checklist to check learners' understanding.



Assessment: Give learners tasks to confirm that they can explore the remainder and factor theorems and apply them in real-life situations.



Learning Activity 6: Using the theorem to find quotient and remainder (including long and synthetic division methods)

Introduction

Using the remainder and factor theorems to find the quotient and remainder of polynomials is important because it allows learners to analyse and simplify complex polynomial expressions

efficiently without always relying on length calculations. By applying these theorems alongside long division or synthetic division, learners can quickly determine whether a linear binomial is a factor, identify the remainder, and compute the quotient, which is essential for solving polynomial equations, factorising higher degree polynomials, and checking solutions. This skill not only strengthens algebraic reasoning and procedural fluency but also provides a foundation for advanced Mathematics, such as calculus, curve sketching, and real-world modeling in physics, engineering, and economics where polynomials often represent relationships that need to be simplified and analysed efficiently.



Problem Posing: A factory models its daily production with the polynomial

$$P(x) = 2x^4 - 7x^3 + 5x^2 + ax - 6$$

Where x represents the number of machines operating, and a is a constant. During testing, engineers discover that when the production model is divided by $(x - 2)$, the remainder is 10.

Questions

- Use the remainder theorem to determine the value of a ,
- Using the value of a , divide $P(x)$ by $(x - 2)$ using synthetic division,
- State the quotient and remainder,
- Verify the result using division algorithm.

Activity 6.1: Discovering the Remainder theorem by long division relay

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial Environment:** Worksheets containing different polynomials.

What to do: Put learners in small manageable groups of mixed abilities. Provide each group with worksheets containing different polynomials. Ensure that each learner solves a part of the long division and passes it on to another learner of the same group until the final solution. The group that finds the final solution first, wins the relay. Provide support to learners where necessary.

Task: Ask learners in groups to solve different polynomial division problems using long division method. Let them present their findings to the class.



Skills to be developed: Collaboration, Communication, and Problem solving.

Activity 6.2: Discovering the Remainder Theorem

Suggested teaching and learning materials to set up the learning environment

- **Artificial environment:** Work cards containing polynomials.

What to do: Put learners in small manageable groups of mixed abilities. Provide each group with work cards containing polynomials (e.g., $P(x) = x^3 - 4x + 1$, and $x - 2$). Ensure that each learner participates in the group activity.

Task: Ask learners to do the following:

1. Divide the polynomial by a given linear expression (e.g., $P(x) = x^3 - 4x + 1$, by $x - 2$).
2. Substitute the value into the polynomial (e.g., $x = 2$, for the case in 1).
3. Compare both answers.



Content tip: Ensure that learners can observe that the remainder when $P(x)$ is divided by $x + a$ is equal to $P(a)$. This leads them to discover the Remainder Theorem.



Skills to be developed: Analytical thinking, Collaboration, Communication, and Problem solving.

Activity 6.3: Discovering Remainder theorem by Synthetic division speed challenge

Suggested teaching and learning materials to set up the learning environment

- **Technological environment:** Worksheets containing questions on synthetic division.
- **Technological environment:** Computers/phones/tablets with a video on synthetic division.

What to do: Put learners in small manageable groups of mixed abilities. Provide them with computers/phones/tablets with downloaded video on synthetic division. Give them instructions on how to find and play the video on the gadgets provided. Thereafter, provide them with worksheets containing questions on synthetic division. Ensure that every learner participates in group work activity.

Task: Ask learners in groups to watch a video on solving polynomials using synthetic division. Let them practice dividing polynomials by the expressions of the form $x - a$ using synthetic division. Have them compare time taken using long division, and using synthetic division. Let them reflect on the following questions:

1. Which method is faster?
2. When is synthetic division applicable?
3. Why does it only work for linear divisors?



Skills to be developed: Analytical thinking, Collaboration, Communication, and Problem solving.

Activity 6.4: Factor theorem investigation

Suggested teaching and learning materials to set up the learning environment

- **Artificial environment:** Work cards containing polynomials and their divisors.

What to do: Put learners in small manageable groups of mixed abilities. Provide each group with work cards. Ensure that every learner participates in group work activity, and provide support to learners where necessary. Let them attempt the problem posing at this stage.

Task: Ask learners to test whether the given linear divisors (e.g., $x - 3, x + 2$) are factors, using substitution. Let them make presentations to the class.



Content tip: Ensure that learners can conclude that if $f(a) = 0$, then $x - a$ is a factor.



Skills to be developed: Collaboration, Communication, Critical thinking, and Problem solving.

Conceptualisation: Make sure that learners can use the theorem to find quotient and remainder correctly, and give them brief notes.

Synthesis: Connect the use of the theorem to find quotient and remainder to real-life situations.

Evaluation: Check if the learners can use the theorem to find quotient and remainder using long and synthetic division methods. Use a checklist to check learners' understanding.



Assessment: Give learners exercises to ensure that they can use the theorem to find quotient and the remainder, and apply them in real-life situations.



Learning Activity 7: Factorising cubic expressions

Introduction

A cubic expression is a polynomial of degree 3, such as $ax^3 + bx^2 + cx + d$ where a, b, c and d are constants and $a \neq 0$. Factorising cubic expressions is important because it allows learners to break down complex polynomials into simpler factors, making them easier to solve, analyse, and apply in real-life situations. By factorising cubic expressions, learners can identify roots or zeros of the polynomial, which is essential for solving equations, sketching graphs and understanding the behavior of functions. It also helps learners apply the remainder and factor theorem efficiently and connects directly to higher level topics like Calculus and Algebra, where finding maxima, minima, or point of intersection often requires solving cubic equations. It also prepares learners to model real-world situations in physics, engineering, and economics where relationships are often represented by cubic functions.



Problem Posing: A teacher wants to arrange learners in a cubical seating plan. The total number of seats S is given by, $S = x^3 - 4x^2 - 7x + 10$. Factorise the cubic function to find how many x learners can sit in each row so that all seats are filled exactly.

Activity 7.1: Factor hunt (Using substitution)

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial Environment:** Worksheets containing cubic expressions to be factorised.

What to do: Put learners in small manageable groups of mixed abilities. Start with the problem posed then provide them with worksheets containing cubic expressions (e.g., $f(x) = x^3 - 4x^2 -$

$7x + 10$). Ensure that all learners participate in group work activity, and provide support to learners where necessary.

Task: Ask learners to do the following:

1. Test possible values of x by trial-and-error method.
2. Find the values of x for which $f(x) = 0$.
3. Identify the factor using the Factor Theorem.



Skills to be developed: Analytical thinking, Collaboration, Communication, Critical thinking, and Problem-solving.

Conceptualisation: Make sure that learners can factorise cubic expressions correctly, and give them brief notes.

Synthesis: Connect factorisation of cubic expressions to real-life situations.

Evaluation: Check if the learners can factorise cubic expressions correctly. Use a checklist to check learners' understanding.



Assessment: Give learners tasks to confirm that they can factorise cubic expressions correctly and apply them in real-life situations.



Learning Activity 8: Solving cubic equations

Introduction

A cubic equation is a polynomial of degree 3, such as $ax^3 + bx^2 + cx + d = 0$, where a, b, c and d are constants and $a \neq 0$. Solving cubic equations is important because it equips learners with the ability to analyse and model real-world situations that involve non-linear relationships. Unlike linear or quadratic equations, cubic equations can describe more complex patterns, such as growth and decay, volume problems, and physical phenomena where the rate of change is not constant. Mastering cubic equations also helps learners identify roots and intercepts, which is essential for graphing functions, optimising quantities and understanding turning points in higher Mathematics.



Problem Posing: A farmer is packing boxes of tomatoes for sale. The total number of boxes (B) depends on the number of boxes per row x and is modelled by

$B = x^3 - 6x^2 + 11x - 6$. If the total number of boxes packed is exactly 0 (meaning all boxes are distributed perfectly without leftovers), find all the possible values of x .

Activity 8.1: Solving cubic equations

Suggested teaching and learning materials to set up the learning environment

- **Artificial environment:** Work cards containing cubic equations to be solved, markers, and manila paper or flip charts.

What to do: Put learners in small manageable groups of mixed abilities. Provide each group with work cards containing cubic equations, manila paper or flip charts, and markers. Ensure that every learner participates in the activity and provide support to learners where necessary.

Task: Ask learners to solve cubic equations and present their findings to the class.



Skills to be developed: Analytical thinking, Collaborations, Communication, and Problem solving.

Conceptualisation: Make sure that learners can solve cubic equations correctly.

Synthesis: Connect solving cubic equations to real-life situations.

Evaluation: Check if the learners can solve cubic equations correctly. Use a checklist to check learners' understanding.



Assessment: Give learners exercises to confirm that they can solve cubic equations and apply them in real-life situations.



Learning Activity 9: Solving identical cubic equations

Introduction

Solving identical cubic equations is important because it helps learners recognise patterns, check consistency, and confirm solutions in both mathematical and real-world contexts. Identical cubic

equations often appear in repeated problems or applications, such as volume calculations, seating arrangements, or parking problems, where the same relationship is modeled in different scenarios. By solving them, learners develop accuracy, confidence, and efficiency in finding roots, understanding multiplicity of solutions, and applying the remainder and factor theorems. It also strengthens algebraic reasoning and problem-solving skills, enabling learners to handle higher degree polynomials in Calculus, Physics, Engineering, and Economics, where verifying identical models is crucial for accuracy and real-life decision making.



Problem Posing: Given the identical cubic equation $(x + 1)^3 = x^3 + 3x^2 + 3x + 1$. Is the statement true for only one value of x , or for all values of x ? How can we prove that it works for all values without testing every number?

Activity 9.1: Solving identical cubic equations

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Worksheets with identical cubic equations.
- ☐ **Technological environment:** Computers/phones/tablets with internet connectivity.

What to do: Put learners in small manageable groups of mixed abilities. Start with the problem posed, then provide each group with worksheets containing identical cubic equations. Provide them with computers/phones/tablets to use when downloading videos on solving identical cubic equations. Guide learners on how to download the videos. You can provide already downloaded videos if need be (in case of places with network challenges).

Task: Ask learners to download and watch the videos on solving identical cubic equations. Let them solve the identical cubic equations on worksheets and present their findings to the class.



Content tip: Ensure that learners solve a variety of questions.



Skills to be developed: Analytical thinking, Collaboration, Communication, and Critical thinking

Conceptualisation: Make sure that learners can solve identical cubic equations correctly, and give them brief notes.

Synthesis: Connect the solving of identical cubic equations to real-life situations.

Evaluation: Check if the learners can solve identical cubic equations correctly. Use a checklist to check learners' understanding.



Assessment: Give learners exercises to confirm that they can solve identical cubic equations and apply them in real-life situations.



Learning Activity 10: Using ICT tools to apply the concepts of the Remainder and Factor Theorems

Introduction

In today's digital world, learners are surrounded by technology, from smartphones to computers and interactive learning platforms. Therefore, using ICT tools to teach the Remainder and Factor Theorems transforms abstract algebraic concepts into interactive, visual, and engaging learning experiences. Through tools like GeoGebra, Desmos, and Excel, learners move beyond rote procedures to exploration, verification, and discovery. This approach supports the Competence Based Curriculum framework by fostering digital literacy, critical thinking, collaboration, creativity, and problem-solving skills. By integrating technology into algebra lessons, educators not only enhance conceptual understanding but also prepare learners with 21st century demands.



Problem Posing: During a mathematics lesson, two groups of learners were asked to investigate whether $(x - 2)$ is a factor of the polynomial $f(x) = x^3 - 4x^2 - 7x + 10$.

One group used manual substitution, while the other used GeoGebra and a scientific calculator. Their answers did not agree. The teacher now asks the class to use ICT tools to determine the correct remainder, verify whether $(x - 2)$ is a factor, and hence factorise the polynomial completely.

Activity 10.1: Verifying the Remainder Theorem using ICT software

Suggested teaching and learning materials to set up the learning environment

- ☐ **Technological environment:** Computers/phones/tablets with software such as GeoGebra, Demos, Symbolab, and a projector for presentations.

What to do: Put learners in small manageable groups of mixed abilities. Start with the problem posed then introduce this activity and explain how ICT software can be applied in the concepts of

Remainder and Factor Theorems. Demonstrate how to use the available software in dealing with concepts of Remainder and Factor Theorems. Have learners also solve given questions on the Remainder and Factor theorems using the software available.

Task: Let each group use the computers/tablets/phones to solve given questions on the Remainder and Factor theorems using the software available. Have them present their findings to the class.



Skills to be developed: Analytical skills, Collaboration, Communication, and Digital literacy

Conceptualisation: Make sure that learners can use ICT tools to apply concepts of the remainder and factor theorems correctly.

Synthesis: Connect the use of ICT tools to apply concepts of the remainder and factor theorems to real-life situations.

Evaluation: Check if the learners can use ICT tools to apply concepts of the remainder and factor theorems correctly. Use a checklist to check learners' understanding.



Assessment: Give learners project-based tasks to confirm that they can use ICT tools to apply concepts of the remainder and factor theorems, and apply them in real-life situations.

Expected Standard: The remainder and factor theorems used correctly in real life context

Summative Assessment Guide:

- ☐ Exploring patterns and relationships in relation to Algebra.
- ☐ Factorising by grouping terms, quadratic expressions, and difference of two squares.
- ☐ Adding, subtracting, multiplying, and dividing algebraic fractions.
- ☐ Simplifying algebraic fractions using Lowest Common Multiples (LCM) and factorisation.
- ☐ Exploring the remainder and factor theorems.
- ☐ Using the theorem to find quotient and remainder (including long and synthetic division methods).

- ☐ Factorising cubic expressions.
- ☐ Solving cubic equations.
- ☐ Solving identical cubic equations.
- ☐ Using ICT tools to apply concepts of the remainder and factor theorems.



Summary:

Key Points Recap:

- ☐ Factorisation by grouping terms is a method used to factor algebraic expressions (usually with four terms). It is done by grouping terms into pairs, then factoring out the common factor from each pair, and factoring out the common binomial factor
- ☐ The difference of two squares is a special algebraic identity: $a^2 - b^2 = (a - b)(a + b)$. It works only when both terms are perfect squares, and there is a minus (–) sign between them
- ☐ Factorisation of algebraic expressions is the process of expressing algebraic expression as a product of two or more simpler expressions (called factors). It is the reverse of expansion. It helps in solving quadratic equations
- ☐ An Algebraic fraction is a fraction that contains variables in the numerator, denominator, or both. These algebraic fractions can be added, subtracted, multiplied or divided.
- ☐ Simplifying using factorisation involves factoring both the numerator and denominator completely, then cancel common factors
- ☐ Remainder Theorem: If a polynomial $f(x)$ is divided by $(x - a)$, then the remainder is given by $f(a)$
- ☐ Factor Theorem: $(x - a)$ is a factor of $f(x)$ if and only if $f(a) = 0$.
- ☐ A cubic polynomial has degree 3
- ☐ ICT tools help verify and apply remainder and factor theorems



Revision exercise

Item 1

A school stores chalk and books in boxes. The number of items is represented by

$$ax + ay + bx + by$$

where a and b represent two storage rooms and x and y represent two types of items.

- (a) Group the terms appropriately,
- (b) Factorise the expression completely,
- (c) State the common binomial factor,
- (d) Explain why grouping is useful in this situation.

Item 2

The function below represents the profit (in thousands of dollars) for a Zambian company;

$$P(x) = x^3 + 2x^2 - 9x - 18$$

- (a) Factorise $P(x)$ completely using factorisation by grouping terms,
- (b) Use the Factor Theorem to verify one of the linear factors,
- (c) Find all the of x for which the profit is zero and interpret their meaning in the business context.

Item 3

A school agriculture club is designing rectangular garden beds. The total area of one garden is represented by;

$$A = x^2 + 5x + 6$$

- (a) Factorise the expression representing the garden area,
- (b) Interpret the factors as possible length and width of the garden,
- (c) The club decides to expand the garden and the new area becomes:

$$A = x^2 - 16$$

Factorise the new expression and explain how the difference of two squares helps

determine the new dimensions.

Item 4

Two water pumps supply water to a school. Pump A supplies water at a rate of $\frac{6}{x-3}$ litres. Pump B supplies water at $\frac{4}{x+2}$ litres per minute.

- (a) Determine the difference in supply rates between Pump A and Pump B,
- (b) Due to a pressure regulator, the difference in supply is multiplied by $\frac{2}{x+12}$, then the result is then divided by $\frac{2}{x+2}$ to determine the final effective rate. Find this final effective rate in a simplified manner.

Item 5

A school is monitoring water usage. The science block uses $\frac{x+2}{x^2-9}$ cubic metres of water per day, while the administration block uses $\frac{x-1}{x^2+x-6}$ cubic metres per day.

- (a) Find the total daily water usage for the school in cubic metres,
- (b) If the school reduces usage by $\frac{2x^2-4x-2}{(x^2-9)(x-2)}$ cubic metres through conservation measures, determine the new daily water usage in a simplified manner.

Item 6

A farmer mixes two types of fertilizer. One contributes nutrients at the rate of $\frac{x^2-1}{x^2-5x+6}$ Kg per hectare, and the other contributes $\frac{x^2-4x+3}{x^2-x-6}$ Kg per hectare.

- (a) Determine the combined nutrient rate per hectare,
- (b) The farmer later uses only half of the mixture. Find the remaining mixture.

Item 7

A packaging company models its weekly profit using the polynomial

$$P(x) = x^3 - 4x^2 - 7x + 10$$

where x represents the number of machines operating.

- (a) Use the remainder theorem to find the remainder when $P(x)$ is divided by $(x - 2)$,

- (b) Use the factor theorem to determine whether $(x - 2)$ is a factor of $P(x)$,
- (c) Factorise $P(x)$ completely,
- (d) Determine the values of x for which the company makes zero profit.

Item 8

A factory models its weekly production using the polynomial

$$P(x) = x^3 + ax^2 + bx + 6$$

where x is the number of operating machines.

When $P(x)$ is divided by $(x - 1)$, the remainder is 10.

When $P(x)$ is divided by $(x - 2)$, the remainder is 12.

- (a) Use the Remainder Theorem to form two simultaneous equations in a and b ,
- (b) Solve the simultaneous equations to determine a and b ,
- (c) Write the complete polynomial $P(x)$,
- (d) Factorise the cubic polynomial completely,
- (e) Solve the equation $P(x) = 0$.

Item 9

The volume of water in a reservoir is modeled by

$$V(x) = x^3 + ax^2 + bx + c$$

where x represents time in hours.

It is known that:

The remainder when $V(x)$ is divided by $(x - 1)$ is 4.

The remainder when $V(x)$ is divided by $(x - 2)$ is 6.

$(x + 1)$ is a factor of $V(x)$.

- (a) Use the Remainder and Factor Theorems to form three equations involving a , b , and c ,
- (b) Solve the systems of equations to determine the constants, a , b and c ,
- (c) Write the full polynomial expression,

- (d) Factorise the cubic polynomial,
- (e) Solve $V(x) = 0$.

Item 10

Engineers use two formulas to represent the volume of water in a storage tank.

Model A

$$V(x) = x^3 + ax^2 + bx - 12$$

Model B

$$V(x) = (x - 3)(x + 2)(x - 2)$$

- (a) Expand model B,
- (b) Since the expressions are identical, determine the values of a and b ,
- (c) Write the full cubic polynomial,
- (d) Solve $V(x) = 0$.

TOPIC 5: MATRICES

Introduction

Overview: Matrices are an important part of Algebra because they provide an organised way of arranging numbers, symbols, or expressions in rows and columns so that information can be represented, analysed, and manipulated efficiently. They are widely used in Mathematics and in everyday life to organise data such as school records, prices of goods, transport schedules, sports scores, election results, inventories, and timetables. Through Matrices, learners are able to compare related information, perform calculations in an orderly manner, and solve practical problems involving many quantities at once.

The study of Matrices enables learners to develop logical thinking, accuracy, and problem-solving skills. It also helps them to appreciate how Mathematics is used to represent real-life situations in compact and meaningful ways. In practical contexts, matrices can be used in business, science,

ICT, engineering, agriculture, and education. Mastery of matrices therefore strengthens analytical skills, mathematical reasoning, and the ability to apply mathematical ideas to everyday situations.

In this topic, the sub-topic is Matrix Operations.

General Competences

- ☐ **Analytical Thinking:** Identify patterns, relationships, and structures in matrices and use them to solve problems.
- ☐ **Collaboration:** Work effectively with others in pair and group activities involving matrix formation, manipulation, and interpretation.
- ☐ **Communication:** Explain matrix concepts, notation, operations, and solutions clearly using appropriate mathematical language and symbols.
- ☐ **Creativity and Innovation:** Apply matrices in representing real-life situations such as timetables, stock records, transport fares, and test scores.
- ☐ **Critical Thinking:** Compare different matrix operations and determine the most suitable method for solving a given problem.
- ☐ **Digital Literacy:** Use calculators, spreadsheets, and other digital tools to represent and compute matrix operations.
- ☐ **Problem Solving:** Apply matrices to solve practical problems involving organised data and systems of equations.



Key Terms/Words/Vocabulary

- ☐ **Matrix:** A rectangular arrangement of numbers or symbols in rows and columns.
- ☐ **Row Matrix:** A matrix with one row.
- ☐ **Column Matrix:** A matrix with one column.
- ☐ **Square Matrix:** A matrix with the same number of rows and columns.
- ☐ **Rectangular Matrix:** A matrix in which the number of rows is different from the number of columns.
- ☐ **Zero Matrix:** A matrix in which all entries are zero.
- ☐ **Equal Matrices:** Matrices that have the same order and corresponding elements equal.
- ☐ **Diagonal Matrix:** A square matrix in which all elements outside the main diagonal are zero.

- ❑ **Identity Matrix:** A square matrix with 1s on the main diagonal and 0s elsewhere.
- ❑ **Order of a Matrix:** The number of rows and columns in a matrix, written as $m \times n$.
- ❑ **Single element matrix:** This is a matrix with only one element. It is also called as a 1×1 matrix.
- ❑ **Element/Entry:** A number or symbol in a matrix.
- ❑ **Transpose of a Matrix:** A new matrix formed by interchanging the rows and columns of a matrix.
- ❑ **Determinant:** A numerical value associated with a square matrix, especially 2×2 or 3×3 , used in solving problems such as finding inverses.
- ❑ **Singular matrix:** A matrix whose determinant is 0.
- ❑ **Inverse of a Matrix:** A matrix which, when multiplied by the original matrix, gives the identity matrix.
- ❑ **Matrix Addition:** Combining two matrices of the same order by adding corresponding elements.
- ❑ **Invertible Matrix:** A matrix with a determinant not equal to 0. It can also be defined as a matrix with an inverse.
- ❑ **Matrix Subtraction:** Subtracting corresponding elements of matrices of the same order.
- ❑ **Scalar Multiplication:** Multiplying every element of a matrix by the same number.
- ❑ **Matrix Multiplication:** Combining two matrices according to the rule of rows by columns, where the number of columns in the first matrix equals the number of rows in the second.
- ❑ **System of Linear Equations:** A set of two or more equations with common variables that can be solved simultaneously.

Sub-topic: Matrix Operations

Introduction

Matrix operations help learners to organise, analyse, and manipulate data in a systematic way. In everyday life, people use tables and arrangements of numbers to represent information such as prices of goods, examination marks, transport routes, attendance records, and stock lists. Matrices make it possible to handle such information efficiently and to carry out operations such as addition, subtraction, multiplication, and solving equations in a structured way.

Through this sub-topic, learners develop the ability to identify different types of matrices, determine their order, perform operations on them, calculate determinants, find inverses, and apply these ideas in practical contexts. This strengthens logical reasoning, computational skills, and the ability to solve real-life problems using organised numerical information.

Specific Competence: Use matrix operations in day to day activities.



Learning Activity 1: Exploring the concept of matrices

Introduction

Exploring the concept of matrices helps learners understand that numbers and data can be organised in rows and columns for easier interpretation and use. In many real-life situations such as recording test scores, shop prices, classroom attendance, transport charges, and sports results, information is naturally arranged in tabular form. This activity introduces learners to matrices as a mathematical way of representing such organised data.



Problem Posing: A teacher records the Mathematics, English, and Science marks for four learners in a table. The head teacher wants the marks arranged in a simpler form so that they can easily compare the performance of learners in different subjects and carry out calculations later.

Questions

- a) How can the information be arranged using rows and columns?
- b) What makes this arrangement a matrix?
- c) How can the entries in the arrangement be identified?
- d) Why is a matrix useful in such a situation?

Activity 1.1: Research on the concept of matrices

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Textbooks, charts, tables, newspapers, record books, price lists, timetables, and worksheets containing organised data.

- ❑ **Technological environment:** Computers/phones/tablets with internet connectivity, and projector for displaying examples of matrices from real-life situations.

What to do: Start with the problem posed. Put learners in small manageable groups of mixed abilities. Provide them with examples of organised information such as school timetables, class attendance sheets, sports score tables, or shop price lists. Guide learners to observe the arrangement of data into rows and columns and help them relate this arrangement to the concept of a matrix. Ask learners to carry out research on the concept of matrices using computers/phones/tablets and encourage them to discuss where they have seen such arrangements in daily life.

Task: Ask learners in groups to identify examples of organised data from their environment and represent them as matrices. Let them carry out research on the concept of matrices and present their findings to the class.



Skills to be developed: Collaboration, Communication, Critical thinking, Creativity and Innovation, Problem solving, and Research skills.

Activity 1.2: Arranging data into matrices

Suggested teaching and learning materials to set up the learning environment

- ❑ **Artificial environment:** Manila papers, markers, counters, and worksheets with tables of data.

What to Do: Present learners with practical information such as prices of mealie meal, sugar, and cooking oil in different shops, or scores obtained by teams in a sports competition. Guide them to arrange the data into rows and columns and identify each entry by position. Help learners understand that each number has a location described by its row and column.

Task: Let learners in groups arrange given sets of information into matrices and identify selected entries using row and column positions. Let them present their work to the class.



Content Tip: Help learners understand that a matrix is not just any table, but a structured mathematical arrangement of entries in rows and columns. Emphasise that position matters, and that each entry can be identified by its location. Encourage learners to move from practical tables to symbolic notation gradually.



Skills to be developed: Analytical Thinking, Collaboration, Communication, and Critical thinking

Conceptualisation: Ensure that the concept of a Matrix as a rectangular arrangement of entries in rows and columns comes out clearly. Give brief notes and examples from daily life such as marks, prices, attendance, and timetables.

Synthesis: Connect the concept of matrices to real-life situations such as school records, shop inventories, transport schedules, sports tables, and agricultural records.

Evaluation: Check whether learners can identify matrix arrangements correctly, distinguish rows from columns, and explain why matrices are useful in organising information.



Assessment: Give learners practical and written tasks in which they organise data into matrices and identify entries by their positions.



Learning Activity 2: Identifying different types of matrices (row, column, square, zero, rectangular, equal, diagonal, identity)

Introduction

Identifying different types of matrices helps learners classify matrices according to their structure. This enables them to recognise similarities and differences between matrices and to prepare for later operations. Learners come to understand that matrices may be row matrices, column matrices, square matrices, rectangular matrices, zero matrices, diagonal matrices, equal matrices, or identity matrices depending on their arrangement and entries.



Problem Posing: A school has different records: one contains a single row of transport costs, another contains a single column of attendance numbers, another records test marks in a square arrangement, and another has only zeros because no books have yet been issued from the library. The teacher wants learners to classify these matrices correctly and it happens that you are among these learners. How can these matrices be grouped according to their form?

Activity 2.1: Classifying matrices from examples

Suggested teaching and learning materials to set up the learning environment

- **Artificial environment:** Flash cards with matrices, and worksheets.

What to do: Start with the problem posed. Put learners in small manageable groups of mixed abilities. Give each group flash cards containing different matrices. Guide learners to observe the arrangement of rows and columns and classify the matrices according to their properties. Offer help where necessary as learners compare different types of matrices.

Task: Ask learners in groups to sort the matrices into categories such as row, column, square, rectangular, zero, diagonal, equal, and identity matrices. Let them explain their answers to the class.



Content Tip: Emphasise that learners should first check the number of rows and columns before naming the type of matrix. Then they should look at special features such as whether all entries are zero or whether the main diagonal contains 1s with all other entries equal to zero.



Skills to be developed: Collaboration, Communication, Critical thinking, and Classification skills.

Conceptualisation: Ensure that learners understand the characteristics of each type of matrix clearly.

Synthesis: Relate the classification of matrices to practical ways of organising information differently depending on the context.

Evaluation: Check whether learners can correctly classify matrices and justify their answers.



Assessment: Provide learners with a mixed set of matrices and ask them to identify the type of each one.



Learning Activity 3: Stating the order of matrices

Introduction

Stating the order of a matrix helps learners describe its size accurately and prepares them for performing Matrix operations correctly. Since many matrix operations depend on the order of the matrices involved, learners need to know how to count rows and columns correctly. This activity strengthens attention to detail and mathematical precision.



Problem Posing: A learner writes a matrix from a classroom attendance record but is unsure of its order. The class now wants to determine the correct way of its order. As a member of the class, help the class to state the correct order.

Activity 3.1: Finding the order of matrices

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Worksheets, manila papers, markers, and charts with examples of matrices of different sizes.

What to do: Place learners in small manageable groups of mixed abilities. Provide each group with charts and guide them to count the rows and columns carefully. Let them write the order in the form $m \times n$. Connect this activity to the problem posed

Task: Ask learners in groups to determine the order of the given matrices and present their answers to the class.



Content Tip: Emphasise that the order of a matrix is written as number of rows by number of columns, not the other way round. Give several examples until learners become confident.



Skills to be developed: Collaboration, Communication, and Presentation.

Conceptualisation: Ensure that learners clearly understand how to determine and write the order of a matrix.

Synthesis: Connect the idea of stating order of a matrix to real-life data arrangement such as learners by subjects, shops by products, or days by activities.

Evaluation: Check whether learners can count rows and columns correctly and state the order accurately.



Assessment: Give learners written exercises containing different matrices and ask them to state the order of each correctly.



Learning Activity 4: Transposing matrices

Introduction

Transposing Matrices helps learners understand that the arrangement of entries in a Matrix can be changed systematically by interchanging rows and columns. This activity develops the ability of learners to observe structure and relationships within Matrices. It also helps them to appreciate how a Matrix can be represented in a different but related form without changing the actual entries. In practical situations, transposition can be used when reorganising information such as changing data recorded by days and subjects into subjects and days, or changing lists of items and prices into prices and items for easier comparison and analysis.



Problem Posing: A class teacher records the marks of learners in three subjects using a matrix in which the rows represent learners and the columns represent subjects. The deputy head teacher wants the same information rearranged so that the rows represent subjects and the columns represent learners.

Questions

- How can the original Matrix be changed to suit the new arrangement?
- What happens to the rows and columns during the change?

Activity 4.1: Investigating the transpose of a Matrix

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Manila papers, markers, flash cards with Matrices, and charts showing rows and columns clearly.
- ☐ **Technological environment:** Projector for displaying matrix examples and animations of row-column interchange, computer/tablet/phone with software such as Microsoft Excel, Google Sheets, and MATLAB or Python (with NumPy) for rotating rows and columns.

What to do: Put learners in small manageable groups of mixed abilities. Start with the problem posed. Provide each group with simple matrices written on flash cards. Guide learners to copy each Matrix and then interchange the rows and columns carefully. Ask learners to use computer/phone/tablet with software to rotate rows and columns and present their findings to the class using a projector for displaying animations. Let them compare the original Matrix with the new one and describe what has changed. Offer help where necessary as learners discuss the meaning of transposition and identify how entries move from one position to another.

Task: Ask learners in groups to find the transpose of given matrices and present their findings to the class. Let them explain how the rows of the original Matrix become the columns of the new matrix.



Content Tip: Help learners understand that to transpose a matrix means to interchange rows and columns. Emphasise that the first row of the original Matrix becomes the first column of the transpose, the second row becomes the second column, and so on. Let learners practice with small matrices first before moving to larger ones. Encourage them to trace the movement of entries carefully so that they develop confidence and accuracy.



Skills to be developed: Collaboration, Communication, Critical thinking, and Digital literacy.

Conceptualisation: Ensure that the concept of the transpose of a matrix comes out clearly. Give brief notes and examples showing how rows become columns and columns become rows.

Synthesis: Relate transposition to real-life situations such as rearranging marks, attendance records, agricultural data, transport tables, and business records so that information can be viewed from a different perspective.

Evaluation: Check whether learners can transpose matrices correctly, explain the movement of entries clearly, and compare the original matrix with its transpose.



Assessment: Give learners written and practical exercises in which they find the transpose of matrices and interpret the meaning of the transpose in real-life contexts.



Learning Activity 5: Performing basic matrix operations on Matrices (addition, subtraction, scalar multiplication and multiplication up to order 3×3)

Introduction

Performing basic Matrix operations helps learners use matrices in a practical and meaningful way. Once learners know how Matrices are formed and classified, they should be able to add, subtract, and multiply matrices by a scalar. They should also begin to understand Matrix multiplication as a structured operation based on rows and columns. These operations allow learners to combine and compare organised data, making Matrices useful in practical situations such as shop records, transport fare tables, school results, budgets, and inventories.

Through this activity, learners strengthen computational accuracy, logical thinking, and problem-solving skills. They also begin to appreciate the conditions under which certain operations are possible, especially that matrices must have the same order for addition and subtraction to be performed.



Problem Posing: Two shops record the number of exercise books, pens, and rulers sold during two days. A teacher wants to combine the sales records, compare the difference in sales, and also calculate how the sales would look if every value were doubled during a promotion. In another case, the teacher wants learners to see how rows and columns can be used to multiply matrices in practical situations.

Question

How can you help the teacher combine the sales records, compare differences in sales, and calculate how the sales would look if the values doubled during a promotion?

Activity 5.1: Performing matrix addition, subtraction, and scalar multiplication

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Manila papers, markers, flash cards with matrices, and worksheets
- ☐ **Technological environment:** Computers/phones/tablets with software which can be used for addition, subtraction, and scalar multiplication of Matrices such as MATLAB where available.

What to do: Put learners in small manageable groups of mixed abilities. Provide each group with pairs of matrices of the same order and guide them to add and subtract corresponding elements. Then provide a scalar and ask learners to multiply each entry of the Matrix by that scalar. Provide the learners with computers/phones/tablets with software and ask them to confirm the solutions they found from the physical calculations. Let them discuss what happens to the values after each operation. Offer help where necessary as learners compare results and explain the meaning of the answers in context.

Task: Ask learners in groups to perform addition, subtraction, and scalar multiplication on given matrices and present their findings to the class. Let them explain each step and interpret the answers using the practical situations provided.



Content Tip: Emphasise that matrix addition and subtraction can only be done when matrices have the same order, because corresponding elements must be combined. Let learners begin with simple examples and practical contexts before moving to abstract exercises. Reinforce accuracy and orderly working at every stage. In scalar multiplication, every entry is multiplied by the same number. Ensure that you give tasks with matrices up to 3×3 .



Skills to be developed: Collaboration, Communication, Critical thinking, Digital literacy, and Problem solving.

Activity 5.2: Exploring matrix multiplication

Suggested teaching and learning materials to set up the learning environment

- ❑ **Artificial environment:** Worksheets with compatible matrices, and charts showing row-by-column multiplication

What to Do: Organise learners in small manageable groups of mixed abilities. Present learners with worksheets. Guide them to multiply matrices using the row-by-column rule. Help learners see that the number of columns in the first Matrix must be equal to the number of rows in the second Matrix. Encourage them to write each step clearly and compare Matrix multiplication with ordinary multiplication.

Task: Let learners in groups multiply compatible Matrices and present their solutions to the class, showing clearly how each entry in the product Matrix is obtained.



Content Tip: For matrix multiplication, learners should follow the row-by-column rule carefully and not assume that multiplication is done entry by entry. Let learners begin with simple examples and practical contexts before moving to abstract exercises. Further, ensure that the multiplication go up to 3×3 matrices.



Skills to be developed: Analytical thinking, Collaboration, Communication, Critical thinking, and Problem solving.

Conceptualisation: Ensure that the concepts of matrix addition, subtraction, scalar multiplication, and matrix multiplication come out clearly. Give brief notes and examples showing the conditions required for each operation and the correct method to use.

Synthesis: Connect basic matrix operations to real-life situations such as combining sales records, comparing transport fares, adjusting prices, analysing school marks, and organising agricultural data.

Evaluation: Check whether learners can identify when matrix operations are possible, perform the operations accurately, and explain their answers clearly.



Assessment: Give learners written and practical exercises requiring them to add, subtract, and multiply matrices by scalars, and to multiply compatible matrices.

Include practical situations so that learners interpret at least some of their answers in context.



Learning Activity 6: Calculating determinants of a 2×2 and a 3×3 matrix

Introduction

Calculating determinants of matrices helps learners to find important numerical values associated with square matrices. Determinants are useful in checking whether a matrix has an inverse and in solving systems of linear equations. Through this activity, learners develop accuracy, logical reasoning, and computational skills. They also begin to appreciate how determinants are applied in practical mathematical procedures and in solving real-life problems that can be represented using matrices.

This activity helps learners move from understanding matrices as arrangements of data to using matrices as tools for analysis and problem solving.



Problem Posing: A farmer keeps records of maize, fertiliser, and seed used in three planting seasons. He arranges the quantities in a square matrix so that he can compare usage across the seasons. How can he use the idea of determinants to make future production decisions?

Activity 6.1: Calculating determinants of 2×2 Matrices

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Flash cards with 2×2 Matrices, worksheets, and charts showing determinant rules.
- ☐ **Technological environment:** Computers/phones/tablets with software such as Wolfram, Symbolab, Mathos AI, Matrix Calculator, and Desmos where available.

What to do: Start with the problem posed. Put learners in small manageable groups of mixed abilities. Provide each group with flash cards containing 2×2 Matrices and guide them to calculate the determinant by multiplying diagonally and subtracting the second product from the first. Ask them to use software to confirm the solutions found after the calculations. Let learners

compare answers and discuss the steps used. Offer help where necessary as they practice more examples.

Task: Ask learners in groups to calculate the determinants of given 2×2 Matrices and verify the solutions using software. Let them explain each step to the class.



Skills to be developed: Collaboration, Communication, Critical thinking, Digital literacy, and Problem solving.

Activity 6.2: Calculating determinants of 3×3 Matrices

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Worksheets with 3×3 Matrices, manila paper, or flip charts, and markers.

What to Do: Put learners in small manageable groups of mixed abilities. Present them with worksheet and guide them through the procedures of finding the determinants. Let them show each step carefully and encourage them to work systematically to avoid errors.

Task: In groups, let learners calculate the determinants of given 3×3 Matrices. Ask them to present their solutions to the class, clearly showing all working.



Content Tip: Emphasise that determinants are only found for square matrices. Ensure that you expose learners to variety of methods of solving determinants. Let learners be exposed to real-life situations where the concept of determinants is used.



Skills to be developed: Analytical thinking, Collaboration, Communication, and Problem solving.

Conceptualisation: Ensure that the idea of a determinant as a single numerical value associated with a square Matrix comes out clearly. Give brief notes for both 2×2 and 3×3 Matrices.

Synthesis: Connect the concept of determinant of 2×2 and 3×3 Matrices to real-life applications.

Evaluation: Check whether learners can calculate determinants of 2×2 and 3×3 Matrices correctly, and explain the procedure they used through a checklist.



Assessment: Give learners written and practical exercises requiring them to calculate determinants of 2×2 and 3×3 Matrices and explain the meaning of their results.



Learning Activity 7: Using the concept of determinants to find the inverse of a 2×2 and 3×3 Matrix

Introduction

Finding inverse matrices using determinants helps learners understand how a square matrix can be reversed in a way that is similar to finding the reciprocal of a number. The inverse of a matrix is important in solving matrix equations and systems of linear equations. Through this activity, learners discover that not every square matrix has an inverse, and that the determinant helps to decide whether the inverse exists.

This activity strengthens the computational ability of learners, attention to detail, and reasoning. It also helps them connect earlier concepts such as square matrices, determinants, and identity matrices. By working with inverse matrices, learners see how different areas of algebra come together to solve practical problems in an organised way.



Problem Posing: A group of learners is given a matrix representing information from a real-life situation in which they were required to calculate the determinant and make decisions. As they begin the task, they discover that some matrices can be inverted while others cannot. The teacher challenges them to investigate the condition that determines whether a matrix has an inverse and to explain how the inverse can be found and used in solving the problem. Under what condition does the given matrix have an inverse?

Activity 7.1: Finding the inverse of a 2×2 matrix using determinants

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Worksheets with 2×2 matrices
- ☐ **Technological environment:** Computers/phones/tablets with software that can be used to find the inverse of a matrix such as Desmos, MATLAB, and Symbolab where available.

What to Do: Put learners in small manageable groups of mixed abilities. Start with the problem posed then present learners with 2×2 matrices whose determinants are non-zero. Guide them step

by step to find the inverse by first calculating the determinant. Ask learners to confirm their solutions by using software such as Desmos, MATLAB, and Symbolab where available.

Task: Let learners in groups find the inverse of given 2×2 matrices and present their solutions to the class. Let them clearly explain condition (s) under which an inverse can be calculated.



Content Tip: Emphasise that only square matrices can have inverses, and even then, the determinant must not be zero. Help learners understand that the inverse of a 2×2 matrix is found through a clear procedure involving the determinant and rearrangement of entries. Encourage learners to check their answers by multiplying the matrix by its inverse to see whether the identity matrix is obtained. This helps them build confidence and accuracy.



Skills to be developed: Analytical thinking, Collaboration, Communication, Critical thinking, and Problem solving.

Activity 7.2: Finding the inverse of a 3×3 matrix using determinants

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Worksheets with 3×3 matrices
- ☐ **Technological environment:** Computers/phones/tablets with software that can be used to find the inverse of a matrix such as Desmos, MATLAB, and Symbolab where available.

What to Do: Put learners in small manageable groups of mixed abilities. Present them with 3×3 matrices whose determinants are non-zero. Guide them step by step to find the inverse by first calculating the determinant. Ask learners to confirm their solutions by using software such as Desmos, MATLAB, and Symbolab where available.

Task: Let learners in groups find the inverse of given 3×3 matrices and present their solutions to the class. Let them clearly explain condition (s) under which an inverse can be calculated.



Content Tip: Expose learners to a variety of methods where the concept of determinant is used to find the inverse of 3×3 matrices. Encourage learners to check their answers by multiplying the matrix by its inverse to see whether the identity matrix is obtained. This helps them build confidence and accuracy.



Skills to be developed: Analytical thinking, Collaboration, Communication, Critical thinking, and Problem solving.

Conceptualisation: Ensure that the concept of using determinants to find the inverse comes out clearly. Give brief notes to consolidate learners' understanding.

Synthesis: Connect the concept of using determinants to find inverse of matrices to real-life situations.

Evaluation: Use a checklist to check if learners can use the concept of determinants to find the inverse of matrices.



Assessment: Give learners tasks requiring them to use determinants to find the inverse of matrices.



Learning Activity 8: Proving matrices using Commutative, Associative and Distributive (CAD) laws and other matrix properties [eg. $[AB]^T = B^T A^T$, $[A^{-1}]^{-1} = A$]

Introduction

Proving matrix properties helps learners to move beyond computation and develop a deeper understanding of how matrices behave under different operations. Through this activity, learners learn that matrix operations follow certain rules and patterns which can be demonstrated using examples and proofs. These properties help learners to work more confidently and accurately when adding, subtracting, or multiplying matrices.

This activity also strengthens logical thinking, reasoning, and the ability to justify mathematical statements. By proving matrix properties, learners begin to appreciate that mathematics is not only about getting answers but also about explaining why methods work. Such understanding is important for further learning in algebra and for solving more complex problems involving Matrices.



Problem Posing: While solving matrix problems, a learner observes that adding two matrices of the same order gives the same answer even when the positions of the two matrices are swapped. However, when the same learner multiplies two matrices, changing the order of multiplication gives a different answer. This makes the class wonder why addition behaves differently from multiplication when working with matrices.

Question

Why did the learner get the same answer during addition when the positions of matrices were swapped but a different answer when the order was changed in multiplication?

Activity 8.1: Proving the Commutative, Associative and Distributive laws

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Worksheets containing matrices of the same order, manila papers, or flip charts, and markers.

What to do: Put learners in small manageable groups of mixed abilities. Start with the problem posed, then provide each group with worksheets and ask them to perform matrix addition in different orders. Guide them to compare the results and deduce the commutative, associative and distributive laws. Let learners discuss what they observe and record their conclusions.

Task: Ask learners in groups to verify properties such as $A + B = B + A$, $(A + B) + C = A + (B + C)$, and $k(A + B) = kA + kB$ using numerical examples. Let them present their work to the class and explain each result.



Content Tip: Ensure that learners can verify whether commutative, associative and distributive laws work for both addition and subtraction or just addition only.



Skills to be developed: Analytical thinking, Collaboration, Communication, Critical thinking, and Problem solving.

Activity 8.2: Exploring properties of matrix multiplication

Suggested teaching and learning materials to set up the learning environment

- **Artificial environment:** Worksheets with compatible Matrices.

What to Do: Put learners in small manageable groups of mixed abilities. Provide learners with worksheets and guide them to compute products such as AB and BA . Help them compare the answers and see that matrix multiplication is not commutative. Then guide learners to verify associative and distributive properties. Encourage them to discuss why some properties hold and why others do not.

Task: Let learners in groups verify that $AB \neq BA$, $A(B + C) = AB + AC$ and $(AB)C = A(BC)$ where possible. Let them present their findings to the class.



Content Tip: Help learners understand that matrix addition behaves much like ordinary addition when the matrices have the same order, but matrix multiplication is different. Emphasise that while matrix multiplication is associative and distributive over addition, it is generally not commutative. Ensure that laws such as $[AB]^T = B^T A^T$, $[A^{-1}]^{-1} = A$, $[A^T]^T = A$, and $(A + B)^T = A^T + B^T$ are proved. Use simple numerical examples so that learners can discover the properties for themselves. Encourage them to explain every step clearly and not just state the results.



Skills to be developed: Analytical thinking, Collaboration, Communication, and Problem solving.

Conceptualisation: Ensure that learners prove the Commutative, Associative, and Distributive and other matrix properties clearly.

Synthesis: Connect proving of the Commutative, Associative, and Distributive and other matrix properties to real-life situations.

Evaluation: Check if learners can prove the Commutative, Associative, and Distributive and other matrix properties clearly.



Assessment: Give learners tasks requiring them to prove the Commutative, Associative, and Distributive and other matrix properties.



Learning Activity 9: Solving systems of linear equations in two and/ or three variables using Matrices

Introduction

Solving systems of linear equations using matrices helps learners apply matrix concepts to practical problem solving. Many real-life situations involve two or more unknown quantities that must be determined simultaneously. Such situations arise in business, farming, transport, science, and education. Matrices provide a clear and organised method for representing and solving such systems.

Through this activity, learners connect earlier ideas such as matrix formation, determinants, inverse matrices, and matrix operations. This strengthens their understanding of algebra as a whole and helps them appreciate the usefulness of matrices in solving problems involving several unknowns at the same time. Learners also develop reasoning, interpretation, and problem-solving skills.



Problem Posing: A shop sells books and pens all in packs. On one day, Chali got 2 packs of books and 3 packs of pens which gave a total cost of K56.00. On another day, Chali got 3 packs of books and 2 packs of pens which gave a total cost of K54.00. Chali wants to know the price of one pack of books and one pack of pens. Help Chali to know the price of one pack of books and one pack of pens.

Activity 9.1: Solving systems of linear equations in two variables by matrix method

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Worksheets containing questions on systems of linear equations in two variables.
- ☐ **Technological environment:** Computers/phones/tables with internet connectivity, or computers/phones/tablets with downloaded video on solving systems of linear equations in two variables by matrix method.

What to Do: Put learners in small manageable groups of mixed abilities. Start with the problem posed then provide learners with computers/phones/tablets with internet connectivity. Also provide them with worksheets. Guide them to download the video. In case there is no internet

connectivity, provide learners with computers/phones/tablets with downloaded videos. Ensure that each learner participates, and offer assistance where necessary.

Task: Let learners in groups download and watch the video on solving systems of linear equations in two variables using matrix method. Ask them to solve the systems of linear equations in two variables on the given worksheets using matrix method. Have them present solutions to the class.



Content Tip: ensure that you expose learners to different matrix methods of solving systems of linear equations in two variables such as Inverse matrix method and Cramer's rule.



Skills to be developed: Collaboration, Communication, Critical thinking, Digital literacy, and Problem solving.

Activity 9.2: Solving systems of linear equations in three variables by matrix method

Suggested teaching and learning materials to set up the learning environment

- ☐ **Artificial environment:** Worksheets containing questions on systems of linear equations in three variables.
- ☐ **Technological environment:** Computers/phones/tables with internet connectivity, or computers/phones/tablets with downloaded video on solving systems of linear equations in three variables by matrix method.

What to Do: Put learners in small manageable groups of mixed abilities. Provide learners with computers/phones/tablets with internet connectivity. Also provide them with worksheets. Guide them to download the video. In case there is no internet connectivity, provide learners with computers/phones/tablets with downloaded videos. Ensure that each learner participates, and offer assistance where necessary.

Task: Let learners in groups download and watch the video on solving systems of linear equations in three variables using matrix method. Ask them to solve the systems of linear equations in three variables on the given worksheets using matrix method. Have them present solutions to the class.



Content Tip: ensure that you expose learners to different matrix methods of solving systems of linear equations in two variables such as Inverse matrix method, Cramer's rule, and Gauss Elimination method.



Skills to be developed: Collaboration, Communication, Critical thinking, Digital literacy, and Problem solving.

Conceptualisation: Ensure that the concept of solving systems of linear equations two and/or three variables by matrix method comes out clearly.

Synthesis: Relate solving systems of linear equations in two and/or three variables using matrices to practical situations such as finding prices of goods, determining quantities in farming, analysing transport charges, and solving classroom-related problems.

Evaluation: Check if learners can solve systems of linear equations in two and/or three variables using various matrix methods.



Assessment: Give learners tasks on solving systems of linear equations in two and/or three variables using matrix method.



Learning Activity 10: Using ICT tools and software to calculate matrix operations and the inverse

Introduction

Using ICT tools and software in matrices helps learners to explore matrix concepts more efficiently and accurately. In modern learning environments, digital tools such as calculators, computers, spreadsheets, and mathematical software can be used to organise matrices, perform operations, calculate determinants, find inverses, and solve systems of equations. This makes the learning of matrices more practical, interactive, and relevant to real-life applications.

Through this activity, learners develop digital literacy alongside mathematical understanding. They also gain confidence in using technology to check manual work, investigate patterns, and solve problems more quickly. This is important because ICT is widely used in education, business, science, engineering, and data analysis, where matrices are often applied.



Problem Posing: A learner has correctly arranged data into matrices but takes too much time when performing matrix operations manually. The teacher suggests using software to carry out the operations and verify the answers. The class now wants to know how ICT tools can be used in working with matrices.

Question

How can the learner use software to perform and verify the matrix operations and the inverse?

Activity 10.1: Calculating matrix operations and the inverse using ICT tools

Suggested teaching and learning materials to set up the learning environment

- **Technological environment:** Computers/phones/tablets with software such as Desmos, Symbolab, and a projector for presentations.

What to do: Put learners in small manageable groups of mixed abilities. Start with the problem posed and then introduce the activity and explain how ICT software can be applied in calculating matrix operations and the inverse. Demonstrate how to use the available software in dealing with concepts of matrix operations and the inverse. Have learners also solve given questions on solving matrix operations and the inverse using the software available.

Task: Let each group use the computers/tablets/phones to solve given questions on matrix operation and the inverse using the software available. Have them present their findings to the class.



Skills to be developed: Analytical skills, Collaboration, Communication, and Digital literacy

Conceptualisation: Make sure that learners can use ICT tools to solve matrix operation and the inverse correctly.

Synthesis: Connect the use of ICT tools to solve matrix operations and the inverse to real-life situations.

Evaluation: Use the checklist to check if the learners can use ICT tools to solve matrix operations and the inverse correctly.



Assessment: Give learners project-based tasks to confirm that they can use ICT tools to solve matrix operations and the inverse problems.

Expected standard: Matrix operations used correctly in day-to-day activities

Summative Assessment Guide:

- ☐ Exploring the concept of matrices
- ☐ Identifying different types of matrices (row, column, square, zero, rectangular, equal, diagonal, identity)
- ☐ Stating the order of the matrices
- ☐ Transposing matrices
- ☐ Performing basic matrix operation on Matrices (addition, subtraction, scalar multiplication and multiplication up to order 3×3)
- ☐ Calculating determinant of a 2×2 and 3×3 matrix
- ☐ Using the concept of determinants to find the inverse of a 2×2 and 3×3 matrix
- ☐ Proving matrices using Commutative, Associative and Distributive (CAD) laws and other matrix properties [*eg.* $[AB]^T = B^T A^T$, $[A^{-1}]^{-1} = A$]
- ☐ Solving systems of linear equations in two and/ or three variables using matrices
- ☐ Using ICT tools and software to calculate matrix operations and the inverse



Summary

Key points Recap

- ☐ A matrix is a rectangular arrangement of numbers or symbols in rows and columns.
- ☐ Matrices are used to organise, represent, and analyse information in everyday life.
- ☐ The horizontal arrangement in a matrix is called a row, while the vertical arrangement is called a column.
- ☐ The order of a matrix is written as the number of rows by the number of columns.
- ☐ Types of matrices include row matrix, column matrix, square matrix, rectangular matrix, zero matrix, diagonal matrix, equal matrix, and identity matrix.
- ☐ A row matrix has only one row, while a column matrix has only one column.

- ☐ A square matrix has the same number of rows and columns, while a rectangular matrix has a different number of rows and columns.
- ☐ A zero matrix has all its elements equal to zero.
- ☐ A diagonal matrix is a square matrix in which all elements outside the main diagonal are zero.
- ☐ An identity matrix is a square matrix with 1s on the main diagonal and 0s elsewhere.
- ☐ Equal matrices have the same order and corresponding elements equal.
- ☐ The transpose of a matrix is found by interchanging rows and columns.
- ☐ Matrix addition and subtraction can only be performed on matrices of the same order.
- ☐ Matrix addition and subtraction are carried out by adding or subtracting corresponding elements.
- ☐ Scalar multiplication is done by multiplying every element of a matrix by the same number.
- ☐ Matrix multiplication is carried out using the row by column rule.
- ☐ Matrix multiplication is only possible when the number of columns in the first matrix is equal to the number of rows in the second matrix.
- ☐ A determinant is a numerical value associated with a square matrix.
- ☐ The determinant of a 2×2 matrix is found by multiplying diagonally and subtracting.
- ☐ The determinant of a 3×3 matrix is found using a systematic expansion method.
- ☐ Only square matrices can have determinants and inverses.
- ☐ A matrix has an inverse only if its determinant is not zero.
- ☐ The inverse of a matrix is a matrix which, when multiplied by the original matrix, gives the identity matrix.
- ☐ Matrix properties help learners understand how matrices behave under addition, scalar multiplication, and multiplication.
- ☐ Matrix addition is commutative and associative, while matrix multiplication is generally not commutative.
- ☐ Matrix multiplication is associative and distributive over addition.
- ☐ Systems of linear equations can be written in matrix form and solved using inverse matrices.

- ☐ Matrices help learners develop problem-solving, critical thinking, reasoning, accuracy, and interpretation skills.
- ☐ ICT tools such as calculators, spreadsheets, Excel, GeoGebra, and other mathematical software can be used to explore and check matrix operations.



Revision Exercise

Item 1

During break time, the school tuck-shop sold 20 buns, 15 fritters, and 10 bottles of water on Monday and on Tuesday, 25 buns, 18 fritters and 12 bottles of water were sold.

- a) Write a matrix where the first row represents Monday and the second row represents Tuesday, while the columns represent buns, fritters, and bottles of water respectively.
- b) Write the number of fritters sold on Tuesday.
- c) State the order of matrix.
- d) What does the element in the first row and third column represent?
- e) Find the total number of buns sold on both days.
- f) Find the total number of all items sold on Monday.
- g) Write the transpose of matrix.
- h) Using the transpose, state what the first row of A^T represent.

Item 2

Three Form 2 learners at a certain school, Jane, Ruth, and Peter, wrote a Mathematics II and Chemistry tests. Jane got 12 in Mathematics II and 14 in Chemistry, Ruth got 15 in Mathematics II and 10 in Chemistry whereas Peter got 9 in Mathematics II and 13 in Chemistry.

- a) Come up with a matrix where rows represent Jane, Ruth and Peter respectively, and the columns represent Mathematics II and Chemistry respectively.
- b) State the score obtained by Ruth in Mathematics II.
- c) State the score obtained by Peter in Chemistry.

- d) What is the order of the matrix?
- e) Which learner obtained the highest total score? Show your working.
- f) Find the difference between Ruth's Mathematics II score and Peter's Mathematics II score.

Item 3

A farming club applied fertiliser to two gardens over three weeks. In week one, Garden A applied 4 bags of fertilizer, in week two, 6 bags and in week three, 5 bags of fertilizer were applied. Garden B applied 3 bags in week one, 7 bags in week two and 4 bags in week three. where the first row represents Garden A, the second row represents Garden B, and the columns represent Week 1, Week 2, and Week 3 respectively.

- a) Write down the matrix if, the first row represents Garden A, the second row represents Garden B, and the columns represent Week 1, Week 2, and Week 3 respectively.
- b) How many bags were applied to Garden B in Week 2?
- c) Find the total number of bags applied to Garden A.
- d) If the number of fertilisers applied is tripled, find the total number of bags applied in Week 3.
- e) Which garden received more fertiliser in the three weeks? Show your working.

Item 4

The school library lends books to the learners fortnightly. The librarian recorded the number of Mathematics, English, and Science books borrowed by Form 1 learners. In Week 1, 6 borrowed Mathematic, 4 borrowed English and 5 borrowed Science while in Week 2, 8 borrowed Mathematic, 3 borrowed English and 7 borrowed Science.

The number of the same books borrowed by Form 2 learners were as follows; in Week 1, 5 borrowed Mathematic, 2 borrowed English and 4 borrowed Science while in Week 2, 7 borrowed Mathematic, 4 borrowed English and 6 borrowed Science.

Task:

Considering weeks as rows and number of books borrowed as columns:

- a) Construct matrices using the information given.
- b) Find the total number of books borrowed of each type by the Form 1s and Form 2s in two weeks in a matrix form.

Item 5

A minibus made two trips in one day. In the first trip, there were 10 adults, 4 children, and 6 luggage units on board. In the second trip, there were 8 adults, 5 children, and 7 luggage units. Because it was a busy day, the driver estimated that the next day each trip would carry twice as many adults, children, and luggage units.

Task:

Considering the rows represent Trip 1 and Trip 2, while the columns represent adults, children, and luggage units respectively

- a) Organise the given information as matrices.
- b) How many children are expected in Trip 2 the next day?

Item 6

A school canteen sells three types of meal packs A, B, and C. On Monday, the canteen sells 5 Packs of A, 4 Packs of B, and 3 Packs of C while on Tuesday, the canteen sells 6 Packs of A, 2 Packs of B, and 4 Packs of C. The meals are priced at 30, 50 and 70 (ZMW) respectively. Mubanga needs help to arrange the information about meal packs as well as their prices in rows and columns, so that he can easily determine the amount to pay.

Task:

Help Mubanga to;

- a) Organise the information in rows and columns clearly stating the order for the information you have organised,
- b) Calculate the total amount he should pay for the meal packs.

Item 7

At a school tuck-shop, two pupils bought books and pens. The first one bought 2 books and a pen while the second bought 3 books and 2 pens. The tuck-shop keeper now wants to present the information about the items bought in a matrix form. Considering the pupils as rows and the items bought as columns.

Task:

Help the tuck-shop keeper to:

- a) Present the given information (Purchase matrix) in a matrix form:
- b) Find a matrix that can be used to work backwards so that we have only one book and one pen.

Item 8

At the market, 2 tomatoes and 1 onion cost K11.00, while 1 tomato and 2 onions cost K10.00. Let the cost of one tomato be x and the cost of one onion be y .

- a) Form two linear equations from the above information.
- b) Hence
 - (i) solve the formed linear equations using matrices.
 - (ii) State the cost of one tomato and one onion.

Item 9

A school tuck-shop sells packs of buns, biscuits, and juice. On Monday, 2 packs of buns, 1 pack of biscuits, and 3 packs of juice were sold at the total cost of K43.00. On Tuesday, 1 pack of buns, 2 packs of biscuits, and 2 packs of juice at the total cost of K34.00. While on Wednesday, 3 packs of buns, 1 pack of biscuits, and 1 pack of juice were sold at the total cost of K35.00.

- a) Find the cost of one pack of buns, one pack of biscuits, and one pack of juice.
- b) Interpret your answers in the context of the tuck-shop.

TOPIC 6: VARIATION

Introduction

Overview: Variation in Mathematics describes the relationship between two quantities and how a change in one quantity affects the other. It helps learners understand how variables are connected and how mathematical models can represent real-life situations. In this topic, learners explore different types of relationships between variables such as direct, inverse, joint, and partial variations. They learn how to express these relationships using mathematical equations, tables and graphs. For example, when two quantities increase or decrease together at a constant rate, they are said to vary directly. When one quantity increases while the other decreases, they vary inversely. Understanding variation helps learners apply Mathematics to real-life situations such as, speed and time relationships, cost and quantity relationships, pressure and volume in science, and other proportional relationships.

Through this topic, learners develop skills in algebraic thinking, problem-solving, graph interpretation, and mathematical communication.

In this topic, the sub-topic is types of variation.

General Competences:

- ☐ **Analytical thinking:** Identify how variables are related, comparing different types of variations, and predict how changes in one variable affects the other.
- ☐ **Collaboration:** Discuss solutions, and working together to solve variation problems.
- ☐ **Communication:** Explaining mathematical relationships
- ☐ **Creativity and innovation:** Modelling real-life situations using variation equations and graphs.
- ☐ **Critical thinking:** Analysing relationships between variables and determine the type or variation involved.
- ☐ **Digital literacy:** Using calculators, graphing tools, or digital platforms to represent and analyse variations.
- ☐ **Problem solving:** Solve problems involving different types of variations.



Terms /Words/Vocabulary:

- ☐ **Variation:** A relationship between two or more variables where a change in one affects the other.
- ☐ **Variable:** A symbol (usually a letter) that represents a quantity that can change.
- ☐ **Constant of Variation:** A fixed number that relates the variables in a variation equation.
- ☐ **Direct Variation:** When two quantities increase or decrease together in the same proportion.
- ☐ **Inverse Variation:** When one quantity increases while the other decreases proportionally.
- ☐ **Joint Variation:** When one variable varies directly with two or more variables.
- ☐ **Partial Variation:** A relationship between variables where one variable changes partly in direct proportion to another variable and partly as a constant (fixed amount).
- ☐ **Proportional Relationships:** A relationship where two quantities maintain a constant ratio.
- ☐ **Equation of Variation:** A mathematical expression that represents the relationship between variables.
- ☐ **Graph of Variation:** A visual representation of the relationship between variables on a coordinate plane.

Sub-topic: Types of Variation

Introduction

Types of variation help learners understand how quantities change in relation to each other. This knowledge is important for solving practical problems in everyday life and for developing strong mathematical skills. The main types of variation are direct, inverse, joint, and partial variations. These are widely used in real-life situations such as, wages and hours worked, speed and travel time, number of workers and time taken to complete a job, force in physics, intensity of light, and many others.

Specific Competence: Apply the concept of variation in daily contexts.



Learning Activity 1: Exploring related proportions in the environment such as age and height, water temperature and depth and so on

Introduction

In everyday life, many quantities are related and change together. Understanding how these quantities are connected helps learners make sense of the world around them. In mathematics, these relationships are often studied through proportions and variations, where one quantity changes depending on another. For example, as children grow older, their height generally increases, the temperature of water may change with depth, and the length of a shadow changes depending on the position of the sun during the day. Through exploring related proportions in the environment, learners develop important competences such as critical thinking, problem solving, collaboration, observation, and data interpretation. This approach not only strengthens their mathematical understanding but also helps them appreciate how mathematics is applied in daily life.



Problem Posing: Why are older learners usually taller than younger ones? Does age always increase with height in the same proportion?

Activity 1.1: Research on related proportions in the environment

Suggested teaching and learning materials to set up learning environment.

- ☐ **Artificial environment:** Books with content on related proportions in the environment or any other research materials with content on the same.
- ☐ **Technological environment:** Computers/phones/tablets with internet connectivity, and a projector for presentations.

What to do: Start with the problem posed. Put learners in small manageable groups of mixed abilities. Provide them with books or computers/phones/tablets to use when researching. Ensure that every learner participates in group activity. Provide support to learners where necessary.

Task: Ask learners in their respective groups to research on related proportions in the environment and make presentations to the class.



Skills to be developed: Collaboration, Communication, Digital literacy, and Research.

Activity 1.2: Age and height investigation

Suggested teaching and learning materials to set up learning environment.

- ☐ **Natural environment:** Learners
- ☐ **Artificial environment:** Measuring tape, plain papers

What to do: Connect the problem posed to this activity. Put learners in small manageable groups of mixed abilities. Provide them with a measuring tape and plain papers. Ensure that every learner participates in the activity, and provide support to learners where necessary.

Task: Ask learners to collect data on age and height from classrooms or learners from different grades. Let them measure height using a measuring tape and record the ages. Let them create a table of age vs height, plot the data on a graph, and discuss whether the relationship shows proportional growth.



Skills to be developed: Analytical thinking, Collaboration, Communication, Data organisation, and Measuring and recording.



Content Tip: Let learners notice that while height generally increases with age during childhood, it does not increase at a constant proportion, leading to discussion about real-life relationships between variables.

Conceptualisation: Make sure that learners can explore related proportions in the environment such as age and height, water temperature and depth, and so on.

Synthesis: Connect exploring of related proportions in the environment to real-life situations.

Evaluation: Check if learners can explore related proportions in the environment correctly.



Assessment: Give learners project-based tasks to confirm that they can explore related proportions in the environment, and apply them in real-life situations.

Learning Activity 2: Exploring the concept of Variation.



Introduction

Exploring the concept of variation enables learners to understand how different quantities change in relation to each other in real-life situations. Through hands on activities such as investigating distance and time, number of workers and time taken to complete a task, shadow length and time of day, water flow and time, learners observe and analyse patterns of change.



Problem Posing: A group of learners in a school garden observe the growth of a plant. Each week they measure the height of a plant. Assuming that all the necessary conditions are present.

Questions

- (a) How does the height of the plant vary with time?
- (b) Does the plant grow by the same amount every week?
- (c) Can you predict the plant's height after several weeks?

Activity 2.1: Research on the concept of variation

Suggested teaching and learning materials to set up learning environment.

- ☐ **Artificial environment:** Books with content on variation.
- ☐ **Technological environment:** Computers/phones/tablets with internet connectivity, and a projector for presentations.

What to do: The problem posed can be done over a period of time and connect it to this activity. Put learners in small manageable groups of mixed abilities. Provide them with books or computers/phones/tablets to use when researching. Ensure that every learner participates in group activity. Provide support to learners where necessary.

Task: Ask learners in their respective groups to research on the concept of variation and make presentations to the class.



Skills to be developed: Collaboration, Communication, Digital literacy, and Research.

Conceptualisation: Ensure that the concept of variation comes out clearly, and give brief notes on them.

Synthesis: Connect the concept of variation to real-life situations.

Evaluation: Check if learners can explore the concept of variation correctly. Use a checklist to check learners' understanding.



Assessment: Give learners project-based activities to check their understanding of the concept of variation.



Learning Activity 3: Identifying types of Variations (Direct, Inverse, Joint, and Partial).

Introduction

At this level, four types of variations are covered, namely, direct variation, which shows proportional increase or decrease, Inverse variation, which shows opposite change between variables, Joint variation, which involves dependence on multipole variables, and Partial variation, which combines a constant value with a varying component. Understanding types of variation helps learners to recognise how quantities relate in real-life situations. Through investigation, experimentation, and real-world simulations learners can understand how variables relate, and that helps us to predict outcomes, model-real-world situations, and solve practical problems in fields such as science, economics, and engineering.



Problem Posing: Imagine you are planning a school trip. The cost per learner decreases as more learners join the trip, but the total cost increases depending on the number of days and activities planned. Why does the relationship between these quantities change?

Activity 3.1: Identifying types of variation

Suggested teaching and learning materials to set up learning environment.

- ❑ **Artificial environment:** Worksheets with statements describing different types of variations.

What to do: Connect the problem posed to this activity. Put learners in small manageable groups of mixed abilities. Provide them with worksheets with statements describing different types of variations (e.g., distance increasing as speed increases). Ensure that every learner participates in group activity. Provide support to learners where necessary.

Task: Ask learners in their respective groups to identify and categorise different types of variations (direct, inverse, joint, partial) from statements and present their findings to the class.



Skills to be developed: Analytical thinking, Collaboration, and Communication.

Conceptualisation: Ensure that the concept of identifying types of variations (direct, inverse, joint, and partial) comes out clearly, and give brief notes on them.

Synthesis: Connect the concept of identifying types of variations to real-life situations.

Evaluation: Check if learners can identify types of variations correctly. Use a checklist to check learners' understanding.



Assessment: Give learners project-based activities to check if they can identify types of variations and connect them to real-life situations.



Learning Activity 4: Solving different types of variations

Introduction

Solving problems involving different types of variation helps learners understand how quantities change in relation to each other. It improves problem solving skills, helps explain real-life

relationships used in science (Such as Boyle's Law and Ohm's Law), and builds a strong foundation for advanced mathematics and scientific studies.



Problem Posing: Imagine a group of learners travelling to a sports competition. The total transport cost increases as the distance increases, but the cost per learner decreases when more learners share the bus. At the same time, the amount of food needed depends on both the number of learners and the number of days they stay. What types of variations are present here?

Activity 4.1: Solving questions on direct variation

Suggested teaching and learning materials to set up learning environment.

- ☐ **Artificial Environment:** Worksheets containing questions on direct variation.

What to do: Start this activity with the problem posed. Put learners in small manageable groups of mixed abilities. Provide each group with worksheets containing questions on direct variation. Ensure that every learner participates in group work activity. Provide support to learners where necessary.

Task: Ask learners to solve the questions on the worksheets (on direct variation) and make presentations to the class. Let them proceed as follows:

1. Write the relationship between two variables
2. Find the value of the constant of variation (k)
3. Write down an equation connecting the two variables
4. Solve for the variable in question.



Skills to be developed: Collaboration, Communication, Presentation, and Problem solving.

Activity 4.2: Solving questions on inverse variation

Suggested teaching and learning materials to set up learning environment

- ☐ **Artificial environment:** Worksheets containing questions on inverse variation.

What to do: Put learners in small manageable groups of mixed abilities. Provide each group with worksheets containing questions on inverse variation. Make sure that every learner participates in the activity. Offer support to learners where necessary.

Task: Ask learners to solve the questions on the worksheets (on inverse variation) and make presentations to the class. Let them proceed as follows:

1. Write the relationship between two variables
2. Find the value of the constant of variation (k)
3. Write down an equation connecting the two variables
4. Solve for the variable in question.



Content Tip: Ensure that learners understand the difference between direct and indirect (inverse) variation correctly.



Skills to be developed: Collaboration, Communication, Presentation, and Problem-solving.

Activity 4.3: Solving questions on Joint Variation

Suggested teaching and learning materials to set up learning environment

- ☐ **Artificial environment:** Worksheets containing questions on joint variation.

What to do: Put learners in small manageable groups of mixed abilities. Provide each group with worksheets containing questions on joint variation. Ensure that every learner participates in the group activity. Provide support to learners where necessary.

Task: Ask learners to solve the questions on the worksheets (on joint variation) and make presentations to the class. Let them proceed as follows:

1. Write the relationship among variables
2. Find the value of the constant of variation (k)
3. Write down an equation connecting the three variables
4. Solve for the variables in questions.



Skills to be developed: Collaboration, Communication, Presentation, and Problem-solving.

Activity 4.3: Taxi fare investigation (Partial variation)

Suggested Teaching and learning materials to set up learning environment.

- ☐ **Artificial environment:** Worksheets containing taxi pricing model

What to do: Put learners in small manageable groups of mixed abilities. Provide each group with worksheets containing a taxi pricing model (e.g., Base fare: K50.00; Cost per Km: K30.00)

Task: Ask learners to solve the problem and proceed as follows:

1. Calculate fares for distances such as 2 km, 5 km, and 10 km.
2. Record values in a table
3. Form the equation and solve problems involving distance.

Let them present solution to the class.



Content tip: Ensure that learners come up with an equation: $C = 30d + 50$. Let them know that that relationship represents partial variation. Let learners understand the difference between joint and partial variations.



Skills to be developed: Analytical thinking, Collaboration, Communication, Critical thinking, Financial literacy, and Problem solving.

Conceptualisation: Ensure that the concepts of solving different types of variation comes out clearly, and give brief notes on them.

Synthesis: Connect the concepts of solving different types of variation to real-life situations.

Evaluation: Check if learners can solve different types of variations correctly. Use a checklist to check learners' understanding.



Assessment: Give learners exercises on solving different types of variations and apply them in real-life.



Learning Activity 5: Presenting Variations Graphically

Introduction

Through practical activities, learners discover that variation describes how quantities change in relation to each other. By using graphs, these relationships become easier to understand, interpret, and apply in everyday life such as shopping, travel, and construction. Graphical representation therefore becomes an important mathematical tool for analysing patterns and making predictions.



Problem Posing: Imagine two friends walking to school every day. One walks slowly while the other walks faster. If we recorded time taken and distance covered, how could we show the relationship clearly so that anyone can quickly understand who walks faster?

Activity 5.1: Market price investigation

Suggested teaching and learning materials to set up learning environment.

- ☐ **Artificial environment:** Worksheets, and graph papers

What to do: Put learners in small groups of mixed abilities. Start with the problem posed. Provide learners with worksheets containing a scenario (e.g., Tomatoes cost K30.00 per Kilogram). Ensure that every learner participates in group work activity, and provide support to learners where necessary.

Task: Ask learners in groups to calculate the cost of buying 1 Kg, 2 Kg, 3 Kg, 4 Kg, 5 Kg, and complete the data in tabular form, like the one below:

Kilograms	1	2	3	4	5
Cost (ZMK)	30	60			

Let learners plot cost against kilograms on a graph, analyse the pattern and determine if it represents direct variation.



Content Tip: Let learners discuss what happens to the cost when kilograms increase, if the graph can pass through the origin, and what a straight line represents. Expose learners to the graphs of all the four variations (direct, inverse, joint, and partial).



Skills to be developed: Analytical thinking, Collaboration, Communication, Graphing, Numerical reasoning, and Problem solving.

Activity 5.2: Workers and time investigation

Suggested teaching and learning materials to set up learning environment.

- ☐ **Artificial environment:** Worksheets containing questions on inverse variation, and graph paper.

What to do: Put learners in small manageable groups of mixed abilities. Provide them with worksheets containing question on inverse variation. Ensure that every learner participates in group work activity. Provide support to learners where necessary.

Task: Ask learners to work out the scenario, if a job requires 12 working hours to complete. Let them complete the table like the one below:

Number of Workers (x)	1	2	3	4	6
Time Taken (Hours)	12	6			

Let learners plot time (y) against number of workers (x) on the graph, and join the plotted points smoothly.



Content Tip: Let learners discuss what happens to time when number of workers increases, if the graph forms a straight line or a curve, and why this is called inverse variation.



Skills to be developed: Analytical thinking, Collaboration, Communication, Graph plotting, and Problem solving.

Conceptualisation: Ensure that the concept of presenting variations graphically comes out clearly. Let learners do a variety of activities and give brief notes on them.

Synthesis: Connect the concept of presenting variations graphically to real-life situations.

Evaluation: Check that learners can present variations graphically, and apply them in real-life situations.



Assessment: Give learners exercises on presenting variations graphically.



Learning Activity 6: Illustrating Variations by incorporating multimedia elements such as videos, animations and interactive graphics

Introduction

Allowing learners to illustrate variations using multimedia is important because it improves understanding, promotes engagement, develops key competencies such as digital literacy, critical thinking, and makes mathematics more relevant and interactive.



Problem Posing: Showing a short animation of a car travelling different speeds, and then asking the following questions:

1. If a car moves faster, what happens to the time taken to reach the destination?
2. If the speed doubles, what do you think happens to the time?

Activity 6.1: Watching and interpreting a video demonstration.

Suggested teaching and learning materials to set up learning environment.

- ☐ **Technological environment:** A computer/phones/tablets, and a projector for presentations.

What to do: Start this activity with the problem posed. Put learners in small manageable groups of mixed abilities. Provide them with computers/phones/tablets for them to use when downloading the videos. Ensure that every learner participates in the activity.

Task: Ask learners to download variety of videos on variations. Allow them to observe the different varying relationships shown in the videos. Let learners discuss in pairs what happens when one quantity increases or decreases.



Skills to be developed: Analytical thinking, Collaboration, Communication, Interpretation, and Observation skills.

Activity 6.2: Exploring Interactive Graphs

Suggested Teaching and Learning Materials to Set up Learning Environment.

- ☐ **Technological Environment:** A computer or a smartphone

What to do: Put learners in small manageable groups of mixed abilities. Provide them with multimedia gadgets.

Task: Ask learners to manipulate values of variables (e.g., change of values of x and observe changes in y), using interactive digital graphs or animations. Let them adjust values on the interactive graphs, and record how variables change, as well as identifying whether the variation is direct or inverse.



Skills to be developed: Data analysis, Digital literacy, Graph interpretation, and Logical reasoning.

Activity 6.3: Creating simple visual representation.

Suggested teaching and learning materials to set up learning environment.

- ☐ **Technological environment:** A computer or a smartphone

What to do: Put learners in small manageable groups of mixed abilities.

Task: Ask learners to create charts, sketches, or simple digital slides illustrating variation relationships from the activities. Let them draw graphs or diagrams showing how two variables relate, and present their findings to the class.



Skills to be developed: Communication, Collaboration, Creativity and Innovation, Digital literacy, and Presentation skills.

Conceptualisation: Ensure that the concept of illustrating variations by incorporating multimedia elements such as videos, animations, and interactive graphics comes out clearly. Let learners do a variety of illustrating variations by incorporating multimedia elements such as videos, animations, and interactive graphics

Synthesis: Connect the concept of illustrating variations by incorporating multimedia elements such as videos, animations, and interactive graphics to real-life situation.

Evaluation: Check if learners can illustrate variations by incorporating multimedia elements such as videos, animations, and interactive graphics correctly.



Assessment: Give learners project-based tasks on illustrating variations by incorporating multimedia elements such as videos, animations, and interactive graphics, and link them to real-life situations.

Summative Assessment Guide:

- ☐ Exploring related proportions in the environment such as age and height, water temperature and depth and so on.
- ☐ Exploring the concept of variation.
- ☐ Identifying types of variations (direct, inverse, joint, and partial).
- ☐ Solving different types of variations.
- ☐ Presenting variations graphically.
- ☐ Illustrating variations by incorporating multimedia elements such as videos, animations and interactive graphics.



Summary:

Key Points Recap:

- ☐ Variation explains how quantities are related.
- ☐ Constants of variation help describe the strength of the relationship.
- ☐ Graphs help visualise relationships between variables.
- ☐ Variation is useful in science, business, and everyday life.
- ☐ Direct variation occurs when two variables change in the same direction. If one variable increases, the other also increases proportionally, and if one decreases, the other decreases also. Mathematical form: $y \propto x$; $y = kx$, where k is the constant of variation.
- ☐ Inverse variation occurs when one variable increase while the other decreases. Mathematical form: $y \propto \frac{1}{x}$; $y = \frac{k}{x}$, where k is the constant of variation.
- ☐ Joint variation occurs when one variable varies directly as the product of two or more other variables. Mathematical form: $y \propto xz$; $y = kxz$, where k is the constant of variation.

- ☐ Partial variation occurs when a variable has both a constant part and a part that varies directly with another variable. Mathematical form: $y = kx + C$, where k is the constant of variation, and C is a constant term.

Expected Standard: The concept of variation applied correctly in daily context.



Revision Exercise

Item 1

State the type of variation in each of the following situations:

- (a) The cost of fuel varies with the number of litres bought.
- (b) The time taken to finish a job varies with the number of workers.
- (c) The volume of a box varies with its length and width.
- (d) A taxi fare includes a fixed charge plus a cost depending on a distance travelled.

Item 2

A small factory produces bottled juice. The number of bottles produced varies directly with the number of hours the machine operates. When the machine runs for 4 hours, it produces 320 bottles of juice.

- (a) Write an equation that relates the number of bottles B to the number of hours h ,
- (b) How many bottles will be produced in 7 hours?
- (c) How long will it take to produce 1,200 bottles of juice?

Item 3

A group of workers is hired to paint a school building. The time required to complete the work varies inversely with the number of workers. It takes 12 days for 6 workers to finish the job.

- (a) Write the inverse variation equation relating time t and the number of workers w ,
- (b) How long will the job take if 9 workers are hired?
- (c) How many workers are needed to finish the job in 4 days?

Item 4

The amount of electricity used by a heater varies jointly with the time it is used and the square of the voltage supplied. When the heater runs for 3 hours at 120 V, it consumes 540 units of energy.

- (a) Write the joint variation equation,
- (b) Find the constant of variation,
- (c) How much energy will be used if the heater runs for 5 hours at 150 V?

Item 5

A taxi company charges a fixed booking fee plus a cost that varies directly with the distance travelled. Lubuto pays K45.00 for a 10 km trip and K65.00 for a 20 km trip.

- (a) Write an equation for Lubuto's fare,
- (b) Determine the fixed booking fee,
- (c) How much will a 35 km trip cost Lubuto?

Item 6

The time required to download a large file varies directly with the file size and inversely with the internet speed. A 600 MB file takes 5 minutes to download using a 20 Mbps connection.

- (a) Write the variation equation,
- (b) Determine the constant of variation,
- (c) How long will it take to download a 900 MB file using a 30 Mbps connection?

Item 7

The monthly cost of running a water pump consists of a fixed maintenance cost plus a cost that varies directly with the number of hours the pump operates. If the pump runs 40 hours, the cost is K260.00. If it runs 70 hours, the cost is K380.00.

- (a) Form an equation for the cost,
- (b) Find the fixed cost,
- (c) Calculate the cost if the pump runs 100 hours.

Item 8

A farm's crop yield depends on two factors:

A base yield of 2 tons due to natural soil fertility.

Additional yield that varies jointly with the amount of fertilizer used and the area of land planted.

When 4 bags of fertilizer are used on 2 hectares, the total yield becomes 10 tons.

- (a) Develop an equation representing the total yield,
- (b) Determine the constant of variation,
- (c) Estimate the yield if 6 bags of fertilizer are used on 3 hectares,
- (d) Discuss whether this model would remain accurate for extremely large amounts of fertiliser.

Exemplar Items

Item 1: Scenario

The Production Unit club at a Chifushi Secondary School started a small vegetable farming project. The club grows tomatoes and cabbages, and sells them at the local market. The club noticed that the daily income from selling vegetables depends on the number of vegetable boxes sold. After studying their records, they modeled the income function using the algebraic expression:

$P(x) = x^3 - 6x^2 + 11x - 6$, where $P(x)$ represents the profit in hundreds of Kwacha, and x represents the number of vegetable boxes sold in a day.

The club also keeps records of vegetables harvested each week as can be seen in the table below.

	Tomatoes	Cabbage
Week 1	20	15
Week 2	25	18

The selling price per box of tomatoes and cabbages is K250.00 and K200.00 respectively.

The club also observed that the amount of water used for irrigation varies directly with the number of vegetable beds planted. When 5 beds are planted, the club uses 200 litres of water per day.

Task:

- (a) Determine all the possible number of boxes that must be sold for the club not to make any profit;
- (b) Calculate the total number of vegetable boxes harvested in week 2;
- (c) Determine the club's weekly income from the sale of tomatoes and cabbage; and
- (d) Find the required amount of water in litres if the club plants 8 beds.

Item 2: Scenario

Sibeso started a poultry business where she rears chickens and sells eggs at the local market. She uses mathematics to manage her enterprise. The table below shows her production in the first two weeks.

	Tray of eggs (30 eggs each)	Broken eggs
Week 1	40	5
Week 2	45	3

The selling price for each tray of eggs is K95.00 and the broken eggs have no selling value.

Sibeso needs help to arrange the information about the number of trays produced, eggs broken as well as the selling price for each tray in rows and columns so that she can easily determine how much is expected every week.

Sibeso also created a model to estimate profit depending on the number of trays sold:

$F(x) = x^3 - 7x^2 + 14x - 8$, where x represents the number of trays sold and $F(x)$ represents the profit made. During her investigation, she discovered that there would be no profit from the sale of 1 tray of eggs.

Sibeso also noticed that the amount of chicken feed used varies directly with the number of chickens kept. When she keeps 50 chickens, she uses 100kg of feed in a week.

Task:

Help Sibeso to:

- (a) determine all the possible number (s) of trays that must be sold for her not to make any profit.
- (b) organise the information in rows and columns, clearly stating the order of the information you have organized for the number of trays produced and that of the selling price for each tray respectively.
- (c) find her expected amount in each week after selling eggs.
- (d) determine the week in which she had the higher income. Justify your answer.

- (e) calculate the amount of feed required if she increases the number of chickens to 80.

Item 3: Scenario

Chaona is a farmer in Kabwe rural and supplies maize to a milling company that produces mealie meal for sale in nearby markets including Lusaka and Ndola. She records production, transportation and milling costs using mathematical models to help her plan operations efficiently.

Chaona has three major farms A, B, and C that supply maize to the milling plant. The maize is processed into three products namely Breakfast meal, Roller meal, and Bran. The cost of transporting maize from the farms to the milling plant depends on fuel usage and distance. The transport manager models the cost per tonne using the expression $\frac{2x}{x^2-9} + \frac{3}{x+3} - \frac{1}{x-3}$, where x represents the fuel adjustment factor. She notices that there are instances when the transport manager is having difficulties in calculate the transportation costs. She needs help in finding such instances.

Chaona further models the production efficiency of the milling machine using the equation

$x^3 - 6x^2 + 11x - 6 = 0$, where x represents the number of hours (in units of 10 hours) that the milling machine operates before maintenance is required. She needs help in knowing when the milling machine is to be maintained.

The quantities of products produced from the three farms were as follows: Fam A produced 2 tonnes of breakfast, 1 tonne of roller meal, and 3 tonnes of bran; Farm B produced 1 tonne of breakfast, 2 tonnes of roller meal, and 1 tonne of bran, and yet Farm C produced 3 tonnes of breakfast and 2 tonnes of bran. Chaona wants help in organising this information in rows and columns for easy interpretation.

Chaona sells maize products to three markets, Lusaka, Kabwe, and Ndola. The relationships of the sales in a particular week were as follows:

$$x + y + z = 60,$$

$$2x + y + 3z = 110,$$

$$3x + 2y + z = 100 ,$$

where x, y and z represents tonnes of breakfast, roller meal and bran respectively. She needs help to determine the number of tonnes of each type supplied.

Chaona noticed that the rate of mealie meal production, P varies jointly with the number of milling machines m and the number of working hours. During one week, when 4 machines worked for 6 hours, she produced 720 bags of mealie meal.

Task:

Let the learners help Chaona to:

- (a) Find instance (s) when it would be difficult to calculate transportation costs;
- (b) Determine the time when the milling machine requires maintenance;
- (c) Organise the information in rows and columns, clearly stating the order of the information you have organized for the number of tonnes of breakfast, roller meal and bran produced from the three farms;
- (d) Explain using calculations why the inverse would be useful for her when analysing the information organised in rows and columns in (c) above;
- (e) Determine the number of tonnes of breakfast, roller meal and bran supplied; and
- (f) estimate how many bags would be produced if 5 machines operate for 8 hours. Explain how this information can help her plan production during high demand periods in Lusaka and Ndola.

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